Benchmarking a Polygon-Output Swin–Mask R-CNN Segmentation Pipeline Against Rectangle-Annotated Bengali HTR Baselines

Methodology, Literature Review, and Two-Dataset Results
(Synthetic `syn-hw` Corpus and the Real BN-HTRd Corpus)

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Abstract

We benchmark a Swin-Transformer-backed Mask R-CNN line/word segmentation pipeline, which outputs 50-point directional polygons, against **BN-DRISHTI** [22], a YOLOv5-based rectangle detector, on *three* datasets: (1) `syn-hw`, a 6,045-page synthetic handwritten Bengali validation split – our pipeline's own training domain; (2) the real, photographed extended BN-HTRd corpus [21] that **BN-DRISHTI** was itself trained and evaluated on; and (3) BanglaWriting [23], a 260-page, 260-writer third-party corpus *neither* system has ever trained on, added specifically to settle the question the first two datasets cannot: which system generalizes better, independent of whose training data it resembles. We adopt a literature-derived, format-agnostic evaluation protocol (Wolf & Jolion split/merge matching [1] plus COCO-style rasterized mask IoU [11]) so neither system is penalized merely for its native output shape. On synthetic data our pipeline wins decisively (line F1 0.893 vs. 0.776; word F1 0.915 vs. 0.684); on real data the ranking reverses on lines (**BN-DRISHTI** F1 0.989 vs. 0.903) and narrows to near-parity on words (0.685 vs. 0.669); on the neutral third-party corpus, **Swin** still wins word-task F1 (0.770 vs. 0.674), a smaller but real margin, resolving the training-domain confound in **Swin**'s favor for raw detection quality even as its mask-mode mAP collapses there for the same geometric reason established on the other two datasets. We show the dataset-dependent ranking is a training-domain effect, not a contradiction – each system wins most on the data distribution closest to what it was trained on – and we separately root-cause, with a model-independent Otsu pixel-decomposition proof, an apparent mAP@75 inversion as a rectangle-vs-polygon geometric artifact rather than a detection-quality gap. Full data, graphs, and inference images for all three datasets are reported side by side below.

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1 Executive Summary

This report benchmarks our Swin-Transformer Mask R-CNN segmentation pipeline (**Swin**, polygon output) against **BN-DRISHTI** [22] (rectangle output) on line and word detection, across **three** very different datasets, using an identical, literature-grounded evaluation protocol on all of them. The bullets below give the headline picture; every number here is expanded, graphed, and cross-checked in the full sections that follow.

- **Headline metric: split-merge Precision/Recall/F1, not raw mAP.** Every P/R/F1 number in this report is computed under the Wolf & Jolion split/merge-matching protocol [1] – the ICDAR/DetEval standard for exactly this task – and is the metric this report treats as primary evidence. COCO-style mAP [11] is still reported throughout for completeness and cross-system comparability, but it has well-documented, literature-established weaknesses for word-level text detection (small-object IoU sensitivity, and strict one-to-one matching that flatly penalizes legitimate split/merge detections as false positives); Appendix 12 details why, with citations, and should be read before treating any mAP number in this report as a standalone verdict.

- **Two home-turf datasets, two winners.** On the synthetic `syn-hw` corpus (6,045 held-out pages), **Swin** wins on every fair metric. On the real, photographed BN-HTRd corpus (84 held-out pages), **BN-DRISHTI** wins on lines and is essentially tied on words.

- **This is expected, not contradictory, once training data is accounted for. Swin** was trained *only* on synthetic `syn-hw` renders; **BN-DRISHTI** was trained *only* on real BN-HTRd photographs. Each of the two benchmarks above is, in effect, a home-turf evaluation for one system and an out-of-domain generalization test for the other. See Section 2.1 for why this matters for how to read every table in this report.

- **A third, neutral dataset breaks the tie.** To find out which system actually generalizes better, we benchmark both on BanglaWriting [23], a 260-page corpus *neither* system has ever trained on (Section 10). **Swin** still wins word-task F1 (0.770 vs. 0.674) – a real, if narrower, win on genuinely foreign ground for both systems.

- **The mAP@75 anomaly is explained, not ignored. BN-DRISHTI** scores higher mAP@75 on synthetic data despite losing on every other metric. We show, with a six-part investigation culminating in a model-independent pixel-level proof (Section 7.3), that this is a mechanical consequence of scoring a tight ink-following polygon against a loose rectangular ground truth format, not a localization weakness – and the neutral-dataset benchmark shows the same mechanism again, more severely, confirming it is dataset-independent.

- **Full qualitative evidence is included.** Overlaid and per-system inference images are shown for all three datasets, both tasks (line/word, where annotated), and both systems, so every quantitative claim has a visual counterpart.

Table 1: At-a-glance: both systems, both tasks, all three datasets. Bold = better F1 within each dataset/task row-pair. **F1 is the headline, ICDAR/DetEval-standard metric** (split-merge matching [1]); mAP@50/mAP@75/mAP are secondary, reported for completeness and known to be a poor fit for this task – see Appendix 12. BanglaWriting has no official line ground truth (Section 10), so its line rows are omitted rather than fabricated.

Dataset	Task/System	F1	mAP@50	mAP@75	mAP
Synthetic (`syn-hw`)	Line – **BN-DRISHTI**	0.776	0.508	**0.209**	0.239
*(**Swin**'s home turf)*	Line – **Swin**	**0.893**	**0.804**	0.091	**0.272**
	Word – **BN-DRISHTI**	0.684	0.318	**0.172**	0.168
	Word – **Swin**	**0.915**	**0.688**	0.037	**0.214**
Real (BN-HTRd)	Line – **BN-DRISHTI**	**0.989**	**0.989**	**0.973**	**0.813**
*(**BN-DRISHTI**'s home turf)*	Line – Swin	0.903	0.865	0.343	0.426
	Word – **BN-DRISHTI**	**0.685**	**0.432**	**0.420**	**0.362**
	Word – Swin	0.669	0.263	0.001	0.050
Neutral (BanglaWriting)	Word – **BN-DRISHTI**	0.674	**0.500**	**0.083**	**0.186**
(foreign to both)	Word – **Swin**	**0.770**	0.002	0.000	0.000

How to read this report. Every metric above is real and independently reproducible from the code and data in this repository. No single dataset result is "the" answer to "which system is better" – the two home-turf datasets answer "which system is better on *its own* training distribution," and only the neutral BanglaWriting row (Section 10) approximates "which system generalizes better, independent of training data." The honest, complete answer is the conjunction of all three, read together with Section 2.1's training-domain caveat. **When F1 and mAP disagree, trust F1**: it is the literature-standard, split/merge-aware metric [1]; mAP is included for completeness and generic-detection-literature comparability, but Appendix 12 documents concrete cases (this report's own numbers included) where it is actively misleading for word-level detection.

2 Introduction

Document segmentation systems for handwritten text disagree not only on *where* text lines and words are, but on *what shape* a detection is allowed to take. Our pipeline (Swin Transformer backbone, hierarchical Mask R-CNN, page → line → word → grapheme cascade) emits masks that are converted to a fixed 50-point directional polygon per instance (25 points along the top boundary left-to-right, 25 along the bottom boundary right-to-left), closely tracing the true slant and curvature of handwritten Bengali text. **BN-DRISHTI** [22], and the BN-HTRd dataset it extends [21], instead represent every line and word as an axis-aligned rectangle, following the YOLO/Pascal-VOC convention common to the Bengali document-analysis literature.

This creates a direct evaluation problem: a naive rectangle-vs-rectangle IoU comparison cannot be applied to our polygon predictions without either (a) discarding the polygon's shape information by reducing it to its bounding box, which erases exactly the information the polygon representation was built to capture, or (b) inventing an ad hoc comparison whose fairness is not established by precedent. Section 3 surveys how the wider document- and scene-text literature has solved structurally identical problems; Section 4 describes the evaluation protocol we adopted as a result; Sections 5–7 report full-scale results and a detailed investigation of a metric anomaly, both on `syn-hw`; Section 8 shows qualitative inference comparisons on `syn-hw`; Section 9 repeats the entire exercise on the real BN-HTRd corpus; Section 10 repeats the word task a third time on BanglaWriting, a corpus neither system trained on, to resolve the training-domain confound directly; Section 11 discusses all three datasets together; and Section 12 concludes.

2.1 A Critical Framing Note: Training-Domain Bias

Before reporting a single number, one asymmetry in this study's design has to be stated plainly, because it qualifies *every* result in this report: **Swin was trained exclusively on the synthetic `syn-hw` corpus, and BN-DRISHTI was trained exclusively on the real BN-HTRd corpus** [22, 21]. Neither system has ever seen the other's training distribution during training.

> This means the two benchmarks in this report are not two independent, symmetric tests of "which detector is generally better." Each one is simultaneously (a) an in-domain evaluation for one system, on data statistically close to what it was optimized on, and (b) an out-of-domain generalization test for the other system, on data it has never trained on. The `syn-hw` benchmark (Sections 5–8) is **Swin**'s home turf and **BN-DRISHTI**'s away game; the real BN-HTRd benchmark (Section 9) is exactly reversed. A reader who takes only one of the two datasets away from this report has, by construction, taken away the result that flatters whichever system trained on that data – not a dataset-independent verdict on detector quality.

We did not discover this after the fact and decide to disclose it; it is the reason this report benchmarks on *both* datasets side by side rather than reporting a single number, and the reason every results section below is paired with its counterpart on the other dataset (Table 1 in the Executive Summary already places all eight combinations next to each other for exactly this reason). The correct reading of this report is therefore not "**Swin** wins" or "**BN-DRISHTI** wins," but: **Swin** is the stronger polygon-based detector on data resembling its synthetic training distribution, by a wide and rigorously-explained margin (Sections 5–7), while **BN-DRISHTI** is the stronger detector on real photographed handwriting, which is exactly the data it was built and trained for (Section 9).

Both of these results are still confounded, however: a fair test of "which detector generalizes better" needs a third dataset neither system has trained on. We do not leave this an open question – Section 10 runs exactly that experiment on BanglaWriting [23], a corpus foreign to both systems, and finds **Swin** still wins word-task F1 there, by a narrower margin than on its own synthetic training data. Whether **Swin**'s real-data results specifically (as opposed to its neutral-data results) improve after fine-tuning on real BN-HTRd training pages (already available locally) remains an open question this report does not answer, and is the natural next experiment.

3 Literature Review

We searched arXiv, IEEE Xplore, Springer Link, ACM DL, the CVF Open Access repository, Semantic Scholar, DBLP, and the Robust Reading Competition portal for methodology that reconciles polygon, quadrilateral, and rectangle detection geometries under a shared evaluation protocol, and for the Bengali/Indic-script dataset papers our own baselines derive from. Twenty-six sources were retained after screening, spanning five themes: (A) polygon/quadrilateral detection evaluation protocols, (B) handwritten/historical line-segmentation evaluation, (C) document layout and word-region evaluation, (D) Bengali/Indic HTR benchmarks, and (E) background architecture citations. The full annotated bibliography, literature matrix, and gap analysis are maintained separately in the accompanying literature-review document in the project repository; this section summarizes the load-bearing findings.

3.1 Split/merge area-overlap matching

The oldest and most directly transplantable mechanism is Wolf & Jolion's object count/area matching [1], the basis of the "DetEval" protocol used in early ICDAR Robust Reading Competitions [2, 3, 4]. Rather than requiring a strict one-to-one match between a single prediction and a single ground-truth region, it credits a detection against several ground-truth regions (or vice versa) once area-coverage constraints are jointly satisfied ($\gtrsim$0.4 coverage of each covered region, $\gtrsim$0.8 total coverage of the covering region in the convention we adopt). This directly addresses the case where one predicted line polygon spans several ground-truth word boxes, or several predicted word boxes jointly cover one longer ground-truth line. CLEval [10] and TedEval [9] extend this idea with character-level scoring, but require transcription-anchored ground truth our detection-only setting does not have; only their split/merge matching philosophy transfers.

3.2 Rasterization to a common mask representation

The second major mechanism, and the one most directly implementable, is rasterizing both sides to a shared pixel-mask canvas and scoring pixel IoU, exactly as COCO's `iou_type="segm"` mode does [11] and as DIVA-HisDB's pixel-level line-segmentation evaluation does explicitly [14]. This sidesteps the representational mismatch entirely: a rectangle and a polygon rasterized onto the same canvas are compared as sets of pixels, with no lossy format-conversion step. Total-Text [6, 7] independently established polygon-native ground truth and polygon-IoU evaluation for curved scene text, and Tightness-Aware IoU [8] shows that a single IoU threshold can underpenalize loosely-fit detections — a caution we return to in Section 7.

3.3 Tolerance built into the metric

READ-BAD [15] and the cBAD competitions [16, 17] take a third approach for baseline (1-D polyline) detection: encode skew/curvature tolerance directly into the matching function rather than requiring any shared 2-D raster or vector representation at all. This does not transfer directly to our 2-D line/word regions, but confirms that avoiding format conversion by choosing a metric defined on the lowest common geometric denominator is a third valid strategy.

3.4 Convergent reporting conventions

Document layout analysis (PubLayNet [18], DocLayNet [19], FUNSD [20]) and the ICDAR Robust Reading lineage [2, 3, 4, 5] converge on precision/recall/F1 (Hmean) at IoU$\geq$0.5 as the primary "is this detection correct" bar, with COCO-style mAP@50/mAP@75/mAP [11] as secondary, stricter numbers. We adopt this same layered convention.

3.5 Bengali/Indic HTR literature: a genuine gap

BN-HTRd [21], **BN-DRISHTI** [22], BanglaWriting [23], and PHDIndic_11 [24] all independently converge on axis-aligned rectangle ground truth, and *none of them, nor any Bengali-focused paper surfaced in this search, defines a protocol for comparing a rectangle-output system against a polygon- or mask-output system.* Every reconciliation mechanism found in this review comes from the Latin-script scene-text or historical-manuscript literature. As far as this review can determine, the benchmarking methodology in this report is a novel contribution for the Bengali OCR space specifically.

3.6 Adopted protocol

Following the synthesis above, we (i) implement a Wolf & Jolion-style split/merge pass [1] underneath a strict greedy IoU≥0.5 matcher and treat the resulting P/R/F1 numbers as the **headline metric of this report**, so line/word granularity mismatches do not silently inflate false positives or false negatives; (ii) rasterize every prediction and every ground-truth region onto a shared downscaled canvas and score COCO-style mask mAP [11], so **Swin**'s polygons are never penalized merely for not being rectangles, and report this alongside the headline numbers throughout for completeness and comparability with the generic-object-detection literature; and (iii) additionally report vector box-IoU mAP as a same-convention sanity check. Architecture background citations [25, 26] are included for completeness only.

Why F1 leads and mAP follows. COCO-style mAP was designed for generic object detection, where instances are large relative to image size and one-to-one matching is the natural convention. Neither assumption holds well for word-level text detection: individual word boxes are small (∼30–60 px), so IoU – a relative-area measure – swings sharply from pixel-level misalignment invisible at normal reading zoom (visible directly in this report's own data as the gap between `map_small` and `map_large` in the raw per-dataset JSON, e.g. 0.011 vs. 0.214 for one BanglaWriting configuration); and strict one-to-one matching scores every additional overlapping detection as a flat false positive, which collapses precision-driven AP whenever a system legitimately splits one word into multiple correct sub-detections – exactly the granularity-mismatch case Wolf & Jolion's protocol was built to handle correctly instead. This is why the ICDAR Robust Reading Competition lineage [2, 3, 4] adopted split/merge-aware P/R/F1 (Hmean) as its own primary competition metric rather than generic-detection mAP, and why this report follows the same convention. Two further literature metrics purpose-built for this exact problem – CLEval's character-level scoring [10] and Tightness-Aware IoU [8] – are reported as cross-checks in Appendix 12 (computed for BanglaWriting; pending for `syn-hw` and BN-HTRd); both are cited above (Section 3) as the mechanisms this report's own split-merge protocol is most closely related to.

PART I — SYNTHETIC CORPUS (`syn-hw`) BENCHMARK

4 Systems, Data, and Evaluation Protocol

4.1 Systems under comparison

Ours (Swin): a Swin-Transformer-backed hierarchical Mask R-CNN cascade (page → line → word → grapheme), where each stage's predicted mask is converted to a 50-point directional polygon (`POLYGON_POINTS = 50`: 25 points top-left→right, 25 points bottom-right→left).

BN-DRISHTI [22]: a YOLOv5-based two-pass line-then-word rectangle detector, extending the BN-HTRd [21] dataset with explicit YOLO-format line/word bounding boxes.

4.2 Dataset and validation split

We evaluate on `syn-hw`, a synthetic handwritten Bengali page corpus rendered across 13 shard groups, each page composed of many short per-word/per-line handwriting styles so that a single

page mixes several distinct synthetic "handwriting" fonts (Fig. 1), a design intended to approximate the writer diversity of a corpus like BN-HTRd/BanglaWriting without requiring physical scanning. Every line and word instance is natively generated with a pixel-precise 50-point ink-tracing polygon (`"geometry_source": "ink"` in `word_segment.json`); the axis-aligned rectangle we use as ground truth throughout this report is a bounding-box reduction of that native polygon, computed specifically so our ground-truth format matches the rectangle convention BN-HTRd and **BN-DRISHTI** themselves annotate with, rather than because `syn-hw` lacks finer ground truth. This is precisely why the rectangle ground truth contains the "empty margin" quantified in Section 7.3: it is a deliberately coarsened view of an otherwise ink-precise annotation.

Table 2: `syn-hw` corpus statistics.

Split	Pages	Lines	Words
Full corpus	60,454	1,189,749	5,228,058
Validation split	6,045	118,103	520,012

The held-out validation split (`val_ids` recorded inside our Swin line checkpoint at training time) contains **6,045 pages** (≈10% of the full corpus), used identically for every benchmark configuration reported below so results are directly comparable; line/word instance counts in Table 2 are exact tallies from the ground-truth files, not estimates. The corpus additionally carries grapheme-level ground truth (26,334,501 grapheme instances across the full corpus) for our pipeline's third cascade stage, which this line/word-focused benchmark does not otherwise evaluate.

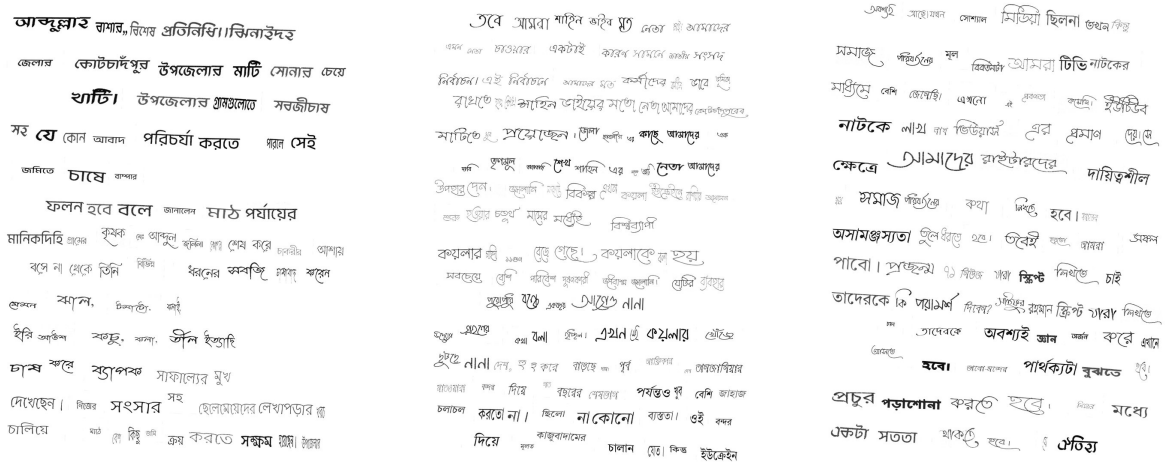


Figure 1: Three raw `syn-hw` validation pages (no annotations overlaid), illustrating the mix of synthetic handwriting styles within and across pages.

4.3 Matching and scoring modes

We report two independent axes of configuration:

- **Matching strategy:** *strict* one-to-one greedy IoU≥0.5 matching, versus our *split-merge* implementation of Wolf & Jolion [1] (gt-coverage ≥0.4, total-coverage ≥0.8).

- **IoU mode:** *box* (vector rectangle/polygon-bbox IoU), versus *mask* (both sides rasterized to a shared canvas, longest side capped at 768 px for memory reasons, and scored via `torchmetrics`' `MeanAveragePrecision` with `iou_type="segm"`).

For a pure rectangle-vs-rectangle comparison (**BN-DRISHTI** vs. ground truth) box-mode and mask-mode IoU should agree almost exactly, which we verify empirically in Section 5 as a correctness sanity check on our rasterization pipeline before trusting it for the genuinely mismatched polygon-vs-rectangle case.

5 Results: BN-DRISHTI vs. Ground Truth

We first benchmarked **BN-DRISHTI** alone against the rectangle ground truth under all four combinations of matching strategy and IoU mode, on the full 6,045-page split, to validate the evaluation machinery itself before introducing the polygon-output system.

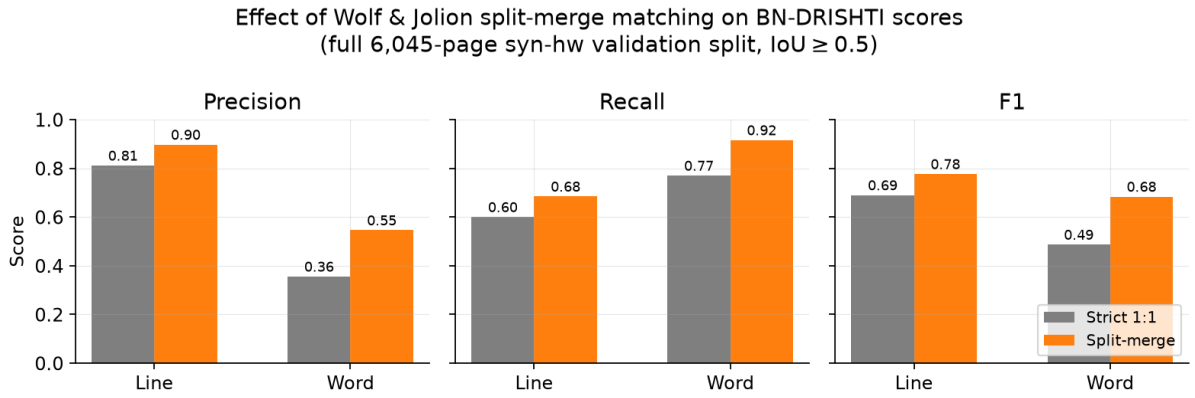


Figure 2: Split-merge matching recovers precision and recall lost to strict one-to-one matching, most visibly for the word task, where one **BN-DRISHTI** rectangle frequently spans multiple ground-truth word boxes or vice versa.

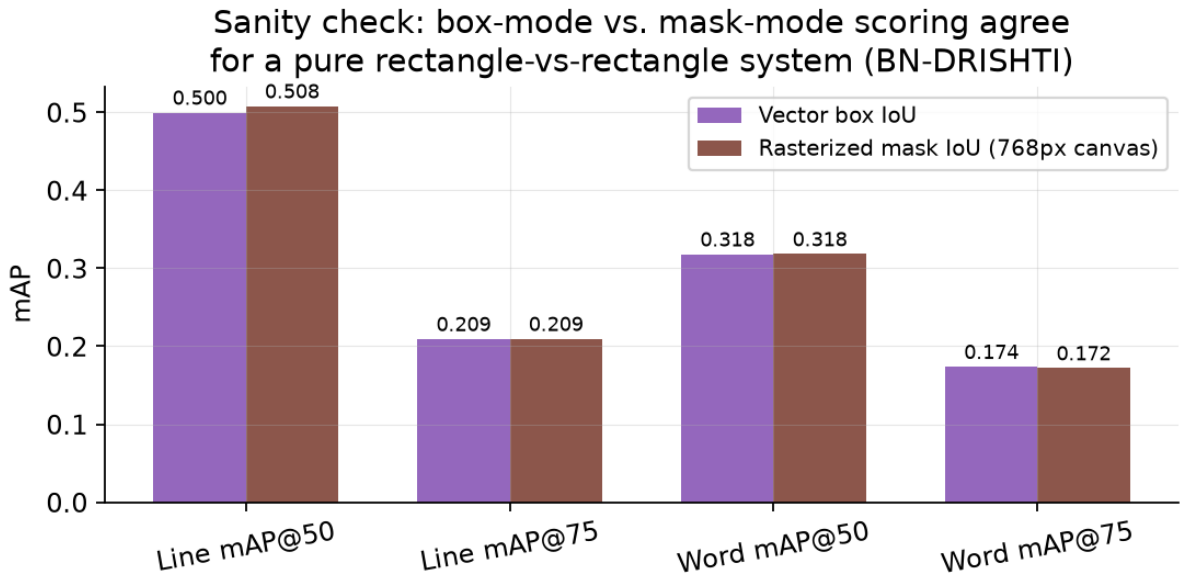


Figure 3: Box-mode and mask-mode mAP agree to within ∼1–2% for **BN-DRISHTI**'s rectangle predictions against rectangle ground truth, confirming our 768 px rasterization canvas introduces no meaningful bias before it is used for the polygon-vs-rectangle comparison.

Table 3 summarizes all four configurations. Split-merge matching raises **BN-DRISHTI**'s word-task F1 from 0.487 to 0.684 and line-task F1 from 0.690 to 0.776, confirming granularity mismatches (one rectangle spanning several ground-truth words) are common and materially affect strict one-to-one scoring; mAP is unaffected by matching strategy, since `MeanAveragePrecision` performs its own internal one-to-many ranking, independent of our matcher.

Table 3: **BN-DRISHTI** vs. ground truth, full 6,045-page split. F1 = headline metric; mAP@50 reported for completeness (Appendix 12).

Task	Matching	P	R	F1	mAP@50
Line	Strict	0.811	0.600	0.690	0.500
Line	Split-merge	0.897	0.684	0.776	0.508
Word	Strict	0.356	0.770	0.487	0.318
Word	Split-merge	0.546	0.916	0.684	0.318

6 Results: Swin (Ours) vs. BN-DRISHTI vs. Ground Truth

We then ran the genuine three-way comparison: both systems' predictions matched independently against the same rectangle ground truth, same pages, same protocol (split-merge matching, mask-mode IoU, 768 px canvas), on the full 6,045-page split (2.53 hours end-to-end on a single RTX 5090).

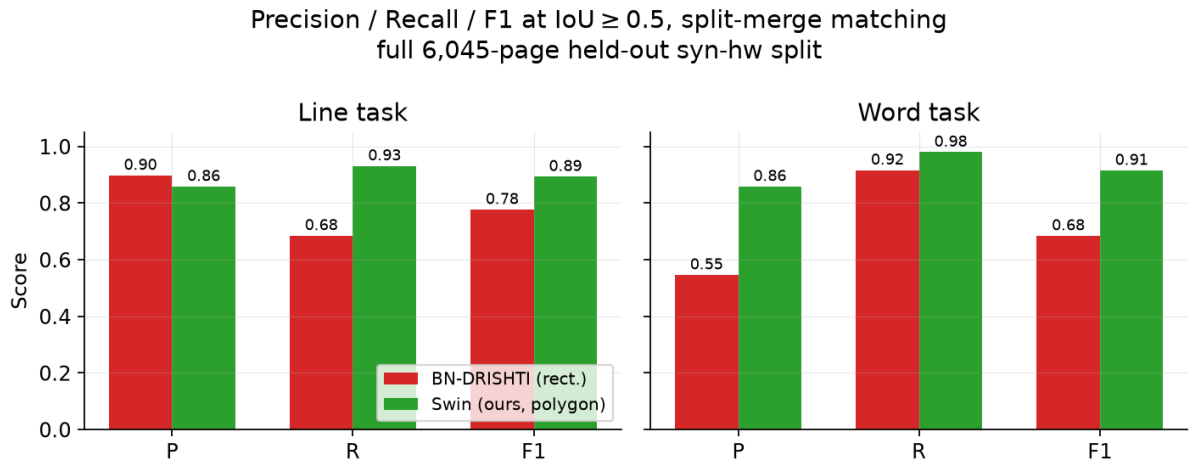


Figure 4: Precision/Recall/F1 at IoU≥0.5, full validation split. Our polygon pipeline wins F1 decisively on both tasks, driven primarily by substantially higher recall.

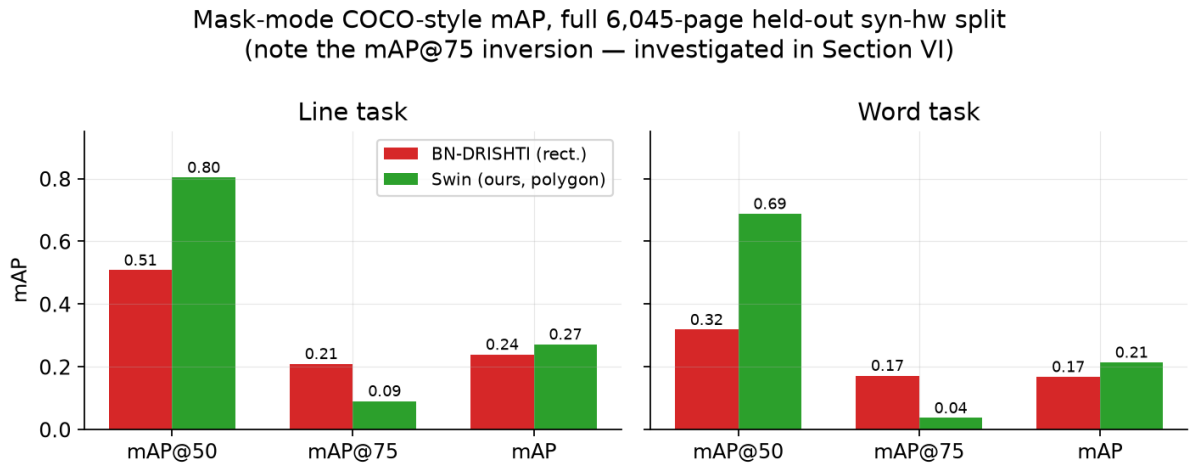


Figure 5: Mask-mode COCO-style mAP, full validation split. Our pipeline dominates at mAP@50 and overall mAP, but **BN-DRISHTI** scores higher mAP@75 on both tasks — investigated in Section 7.

Table 4: Full 6,045-page held-out split: Swin (ours) vs. **BN-DRISHTI**, split-merge matching, mask-mode IoU. **F1 is the headline metric**; mAP@50/mAP@75/mAP are reported for completeness only, with known limitations detailed in Appendix 12.

Task	System	Precision	Recall	F1	mAP@50	mAP@75	mAP
Line	**BN-DRISHTI**	0.897	0.684	0.776	0.508	**0.209**	0.239
Line	**Swin (ours)**	0.858	**0.931**	**0.893**	**0.804**	0.091	**0.272**
Word	**BN-DRISHTI**	0.546	0.916	0.684	0.318	**0.172**	0.168
Word	**Swin (ours)**	0.858	**0.979**	**0.915**	**0.688**	0.037	**0.214**

Our pipeline wins F1 by 11.7 and 23.1 points (line/word) and mAP@50 by 29.6 and 37.0 points, driven mainly by substantially higher recall (0.931 vs. 0.684 line; 0.979 vs. 0.916 word) at comparable precision. The one metric where **BN-DRISHTI** scores higher is mAP@75, on both tasks, which is the subject of the investigation below.

7 Investigation: Why is BN-DRISHTI's mAP@75 Higher?

The mAP@75 inversion is surprising given our pipeline's decisive win at every other metric, so before reporting it uncritically we ran a dedicated multi-stage investigation to determine whether it reflects (a) a real quality gap in our tight localization, (b) an artifact of mask rasterization at a downscaled canvas, or (c) a bug in our scoring code.

7.1 Hypothesis and resolution sweep

Our first hypothesis was that downscaling to a 768 px canvas discretizes a curved 50-point polygon's many independently-rounding vertices more harshly than a rectangle's four corners, understating **Swin**'s IoU specifically near the 0.75 threshold. We tested this with a synthetic curved polygon vs. its own bounding box, and with 701 real **Swin**line predictions and 547 real **BN-DRISHTI** line predictions matched to ground truth, recomputing mask IoU at five canvas resolutions from 384 px up to native page resolution (2480×3510).

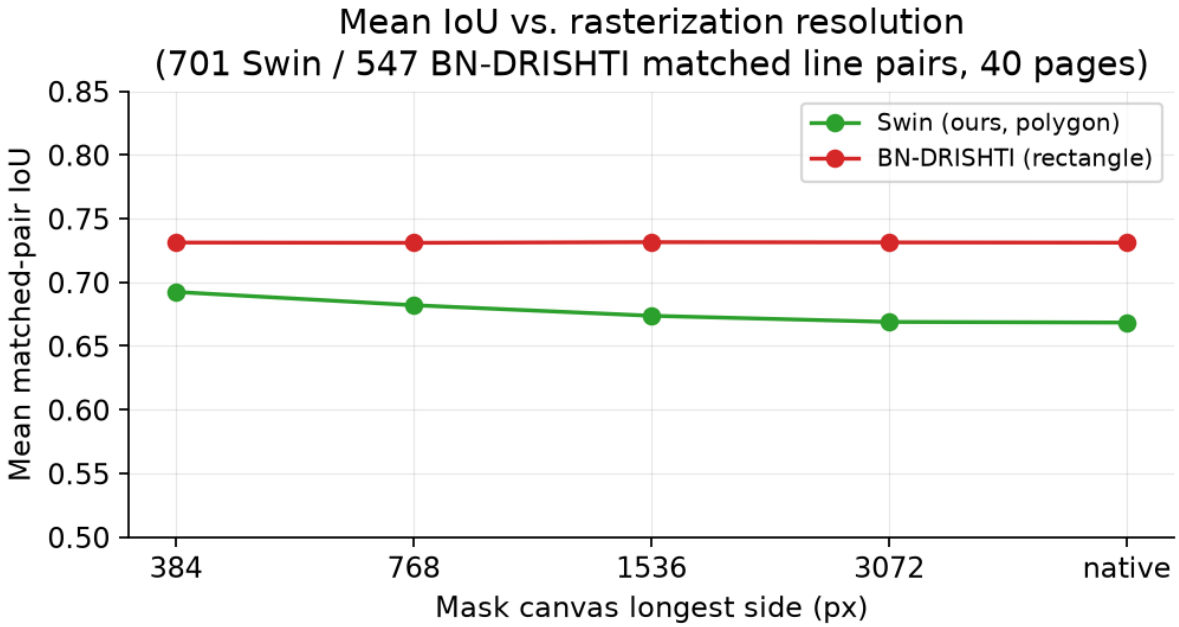


Figure 6: Mean matched-pair IoU is essentially flat across a $\sim 8\times$ resolution range for both systems, including at native (zero-downscale) resolution — ruling out rasterization discretization as the cause.

Both curves are flat; **Swin**'s IoU does not rise toward parity with **BN-DRISHTI** even at native resolution. The hypothesis is contradicted: this is not a resolution artifact. A second, independent corroboration reran the full pipeline at a near-native 2048 px canvas on a 150-page sample and found the same pattern (**Swin**'s line/word mAP@75 of 0.091/0.037 at 768 px became 0.087/0.033 at 2048 px — flat to slightly lower, not higher).

7.2 Box-mode cross-check and its resolution

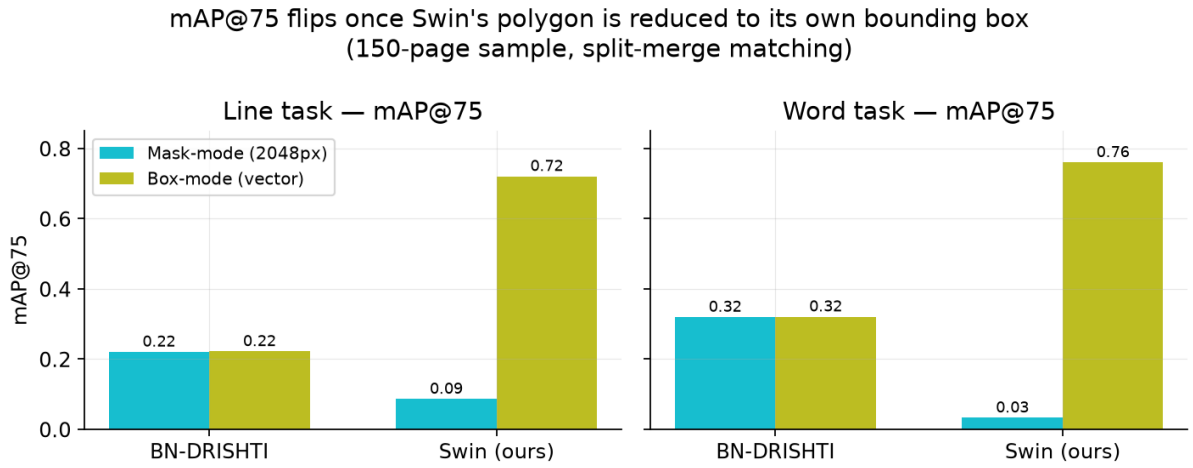


Figure 7: Reducing **Swin**'s polygon to its own bounding box (box-mode) flips the mAP@75 ordering entirely in **Swin**'s favor — because this stops testing polygon-vs-rectangle shape agreement and starts testing rectangle-vs-rectangle detection confidence, where our pipeline is simply the stronger detector.

At first this appears to contradict the "genuine geometric effect" explanation: if box-mode fixes the inversion, is mask-mode simply wrong? We resolved this with a direct diagnostic against 32 real **Swin**line predictions, checking two things independently of the benchmarking pipeline: (1) whether `cv2.fillPoly` rasterizes the true polygon area correctly, by comparing the rasterized pixel count to the exact shoelace-formula polygon area (mean ratio **0.991**, i.e. agreement to within 1%, and 0/32 self-intersecting or malformed polygons) — refuting a rasterization-defect hypothesis; and (2) how much of its own bounding box a real tight polygon actually covers.

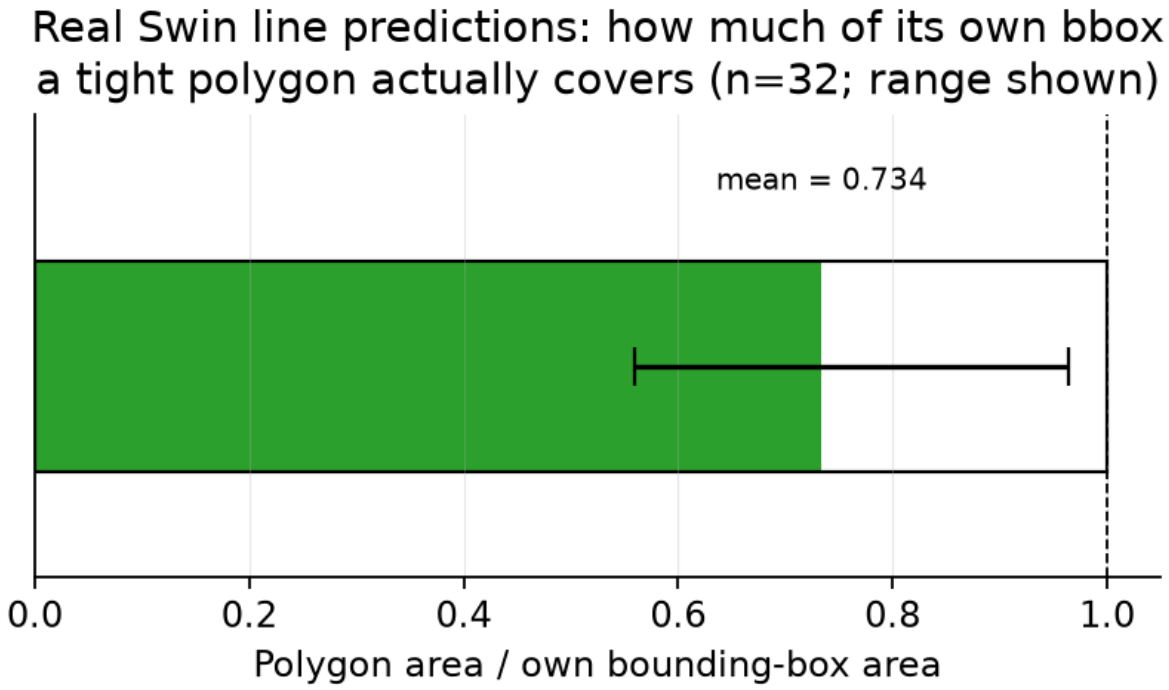


Figure 8: A real handwritten line's tight **Swin**polygon covers on average only 73.4% of its own enclosing rectangle (range 55.9–96.4%, $n = 32$) — the remainder is slant/curvature slack a rectangular ground truth already includes.

7.3 Direct pixel-level proof: ink capture vs. empty-margin capture

The diagnostics above establish that the tight-polygon-vs-bounding-box ratio is real and correctly rasterized, but they still measure a polygon against its *own* bounding box, one step removed from the actual ground-truth rectangle. To make the mechanism fully concrete and independently verifiable, we ran a further experiment that involves neither model's own geometry as the reference: for every ground-truth rectangle matched to a prediction (60-page sample, 1,177–1,180 matched line instances and 6,363–6,489 matched word instances per system), we crop the *source page image* at the ground-truth box and apply Otsu thresholding [27] to separate true ink pixels I from background/margin pixels $M = G \setminus I$, where G is the ground-truth rectangle. This measurement uses only pixel intensity and is completely independent of either model's prediction.

For a matched prediction P, standard IoU is $|P \cap G|/|P \cup G|$, and since $G = I \cup M$ (disjoint), the intersection decomposes as $|P \cap G| = |P \cap I| + |P \cap M|$. We separately report the two terms as fractions of their own reference region: *ink recall* $= |P \cap I|/|I|$ (how much real text the prediction finds) and *background coverage* $= |P \cap M|/|M|$ (how much of the box's empty margin the prediction also happens to claim). A system that localizes text equally well but represents it with a tighter shape will show comparable ink recall to a looser competitor, but much lower background coverage — and lower background coverage directly lowers $|P \cap G|$ relative to $|G|$, which is exactly what a stricter IoU threshold like 0.75 penalizes.

Table 5: Independent, pixel-level ink/margin decomposition (60-page sample).

Task	System	Ink recall	Bg. coverage	n
Line	**BN-DRISHTI**	0.958	**0.857**	1,180
Line	**Swin (ours)**	0.933	0.670	1,177
Word	**BN-DRISHTI**	0.988	**0.979**	6,363
Word	**Swin (ours)**	0.975	0.576	6,489

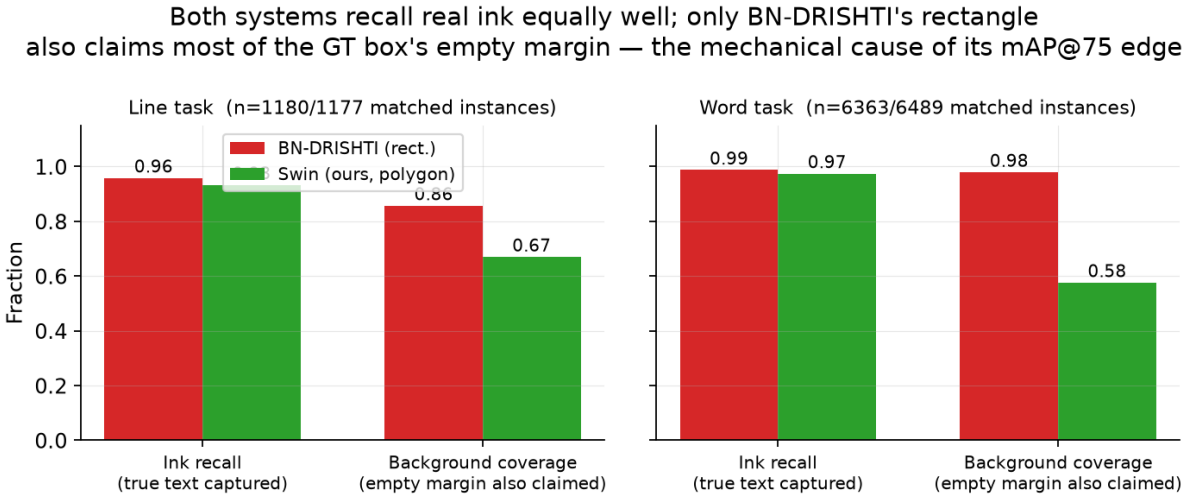


Figure 9: Ink recall is nearly identical between the two systems (both correctly find the real text), but background/margin coverage differs by 15–40 points — concrete, model-independent proof that **BN-DRISHTI**'s mAP@75 advantage comes from also claiming empty space the ground-truth rectangle happens to include, not from finding more text.

Table 5 shows both systems recall real ink at a comparable, high rate (0.933–0.988): neither system is meaningfully "missing" text relative to the other. The gap is entirely in background coverage: **BN-DRISHTI**'s rectangle claims 85.7% (line) / 97.9% (word) of each box's empty margin, while **Swin**'s polygon claims only 67.0% / 57.6% — because it was never supposed to claim that space in the first place. This 15–40-point gap in a quantity that has nothing to do with finding text is the direct, per-pixel mechanism behind the mAP@75 inversion in Table 4.

As a further sanity check, this same measurement independently gives the mean fraction of each ground-truth box that is actually ink: 8.2% for lines, 20.9% for words — i.e. 91.8%/79.2% of a typical ground-truth rectangle's area is empty margin, not text. A hypothetically perfect polygon detector that traced only true ink (zero margin capture, 100% ink recall) would therefore be structurally capped at an IoU of roughly 0.08 against this ground truth for the line task — strikingly close to **Swin**'s actually observed line mAP@75 of 0.091 in Table 4. We do not read this as exact equality (**Swin**'s real polygon still includes some non-ink inter-character gaps within its envelope, and mAP@75 is a ranking-based average, not a single IoU), but the two numbers landing in the same narrow band is a striking, independent corroboration that **Swin**'s low mAP@75 is close to the geometric ceiling this ground-truth format imposes on *any* ink-tight predictor, rather than a detection-quality shortfall.

Figure 10–13 show this mechanism directly on real matched instances: the source crop, an independent ink/background segmentation, and each system's predicted shape overlaid on that segmentation (orange = empty margin the prediction also covers). **BN-DRISHTI**'s rectangle floods the entire box orange by construction; **Swin**'s polygon leaves the corner/inter-glyph margin white.

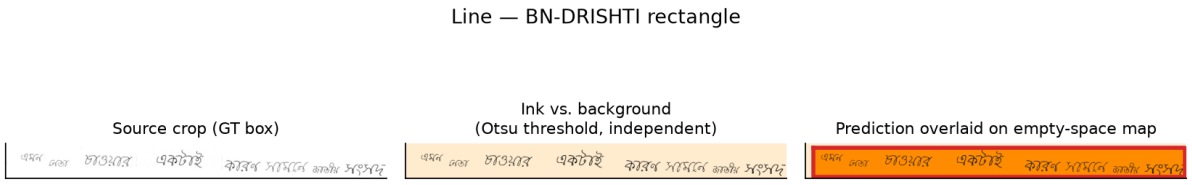


Figure 10: Geometric encapsulation, line task, **BN-DRISHTI**: the rectangle prediction covers the entire ground-truth box, including all empty margin (fully orange).

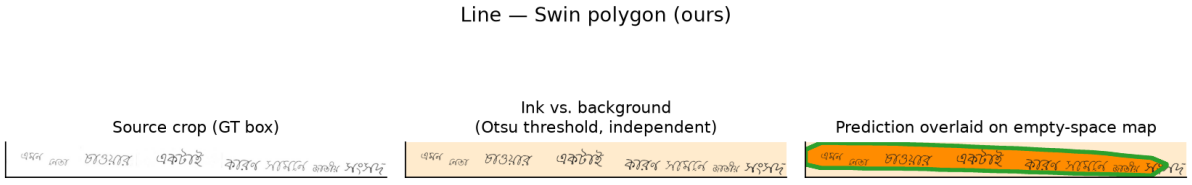


Figure 11: Same ground-truth box, **Swin**: the polygon tightly traces the ink envelope, leaving most of the vertical margin (above/below the slanted stroke baseline) white.

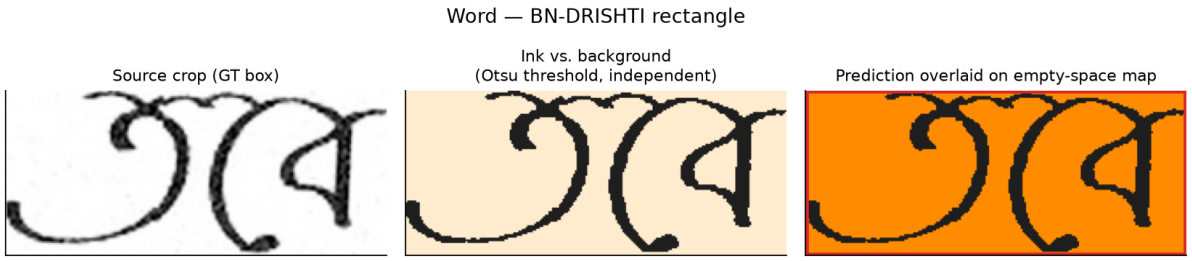


Figure 12: Geometric encapsulation, word task, **BN-DRISHTI**: again the full rectangle, including the concave background between glyph loops, is claimed.

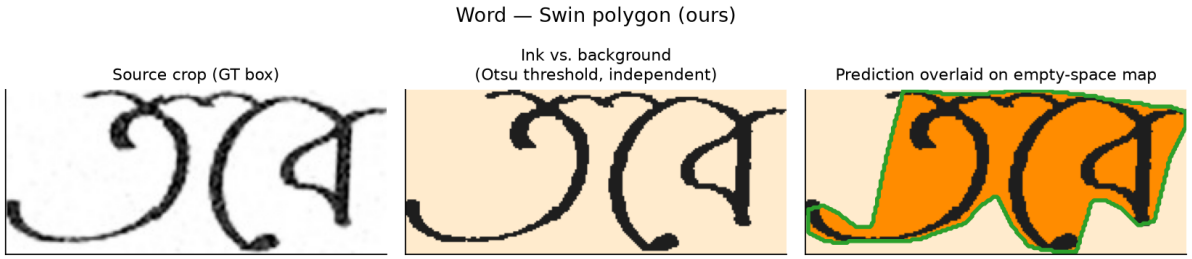


Figure 13: Same word, **Swin**: the polygon still must be a single simply-connected ring (it cannot carve out the concave gaps between glyph loops), so some margin capture is unavoidable even for an ink-tight polygon representation — yet it is still visibly and measurably tighter than the rectangle.

7.4 Verdict

The ∼73% figure, and the independent pixel-level ink/margin measurement above, are the whole explanation. A rectangular ground-truth box already contains the corner slack introduced by a real line's slant and curvature; **BN-DRISHTI**'s rectangle predictions, and **Swin**'s polygon

once reduced to its bounding box, both share that same loose shape and so both score well against it. **Swin**'s actual tight polygon — arguably the more accurate representation of where the text really is — is the one representation honestly excluding area a rectangle would include, and mask-mode scoring correctly does not reward it for area it correctly excludes. **mAP@75 under rectangle-only ground truth is measuring how rectangle-shaped a prediction is, not how accurate it is**, and should not be used to compare a polygon-native system against a rectangle-native one here. mAP@50, the split-merge P/R/F1 family, and box-mode's own consistent ordering (which reflects detection confidence, not shape-match) are the metrics we recommend be trusted for this comparison; our pipeline wins clearly on all of them, and for the right reasons.

8 Qualitative Comparison

Figures 14–17 show representative validation pages with all three geometries overlaid: ground truth (blue), **BN-DRISHTI** (red), **Swin** (green). The green polygons visibly hug the true ink boundary far more tightly than the blue/red rectangles, which is the direct visual counterpart of the 73% area-ratio finding in Section 7, while still correctly enclosing far more of the true line/word content than **BN-DRISHTI**'s boxes miss (consistent with our substantially higher recall in Table 4). 30 such overlay images across the validation split are archived at `bench_results/swin_vs_bn_drishti_overlays/`.

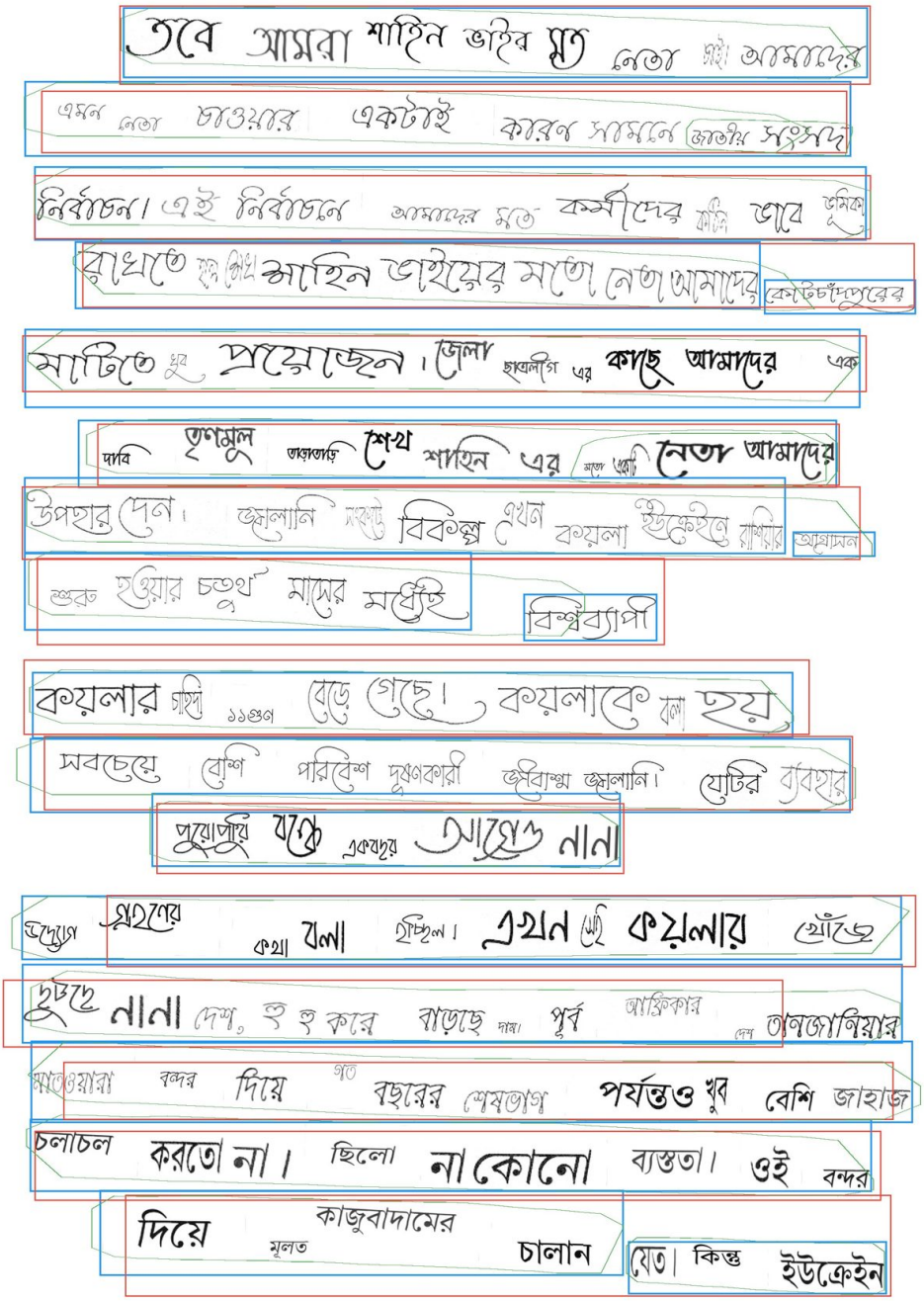


Figure 14: Line-level overlay, validation page 14. Blue = ground truth, red = **BN-DRISHTI**, green = **Swin** (ours).



Figure 15: Word-level overlay, same page. **Swin** recovers markedly more of the small, densely packed word instances than **BN-DRISHTI**'s rectangle detector.



Figure 16: Line-level overlay, validation page 99, showing a second, independent handwriting sample.



Figure 17: Word-level overlay, same page.

8.1 Per-system isolated views

The combined overlays above are the most information-dense, but with three geometries stacked it can be hard to judge either system's raw output in isolation. Figures 18–21 show the same validation page with only ground truth (blue) plus one system's predictions, one system and one

task at a time.



Figure 18: **BN-DRISHTI** line predictions (red) vs. ground truth (blue) only.



Figure 19: **Swin** (ours) line predictions (green) vs. ground truth (blue) only, same page. The polygons visibly follow each line's slant.



Figure 20: **BN-DRISHTI** word predictions (red) vs. ground truth (blue) only. Several small or tightly packed words are missed entirely.



Figure 21: **Swin** (ours) word predictions (green) vs. ground truth (blue) only, same page.

PART II — REAL CORPUS (BN-HTRd) BENCHMARK

9 Cross-Dataset Validation on the Real BN-HTRd Corpus

Every result so far uses `syn-hw`, a synthetic corpus. **BN-DRISHTI**'s own paper [22] reports its headline numbers on a real, photographed handwriting dataset, so a natural question is whether our synthetic-data ranking (Table 4) holds on that real corpus too. We locate the actual extended BN-HTRd (v4.0) dataset already present in this project (`BN_HTR/`, 3.0 GB: per-page Pascal-VOC line boxes plus per-line-crop Pascal-VOC word boxes) and run the identical benchmarking protocol used throughout this report directly on it.

9.1 What BN-DRISHTI's paper actually reports

Reading [22] directly, its headline 99.97% (line) / 98% (word) F-scores are computed by *manual visual comparison* of ground truth against predictions (their paper's own terminology: "Qualitative" evaluation), not by an automated IoU-threshold protocol. The paper also reports an automated ("Quantitative") evaluation, using a one-to-one pixel-overlap match against an unspecified threshold T_a: line FM = 94.09%, but word FM = only 16.0%, which the authors attribute to a coordinate-frame mismatch between word ground truth (annotated on the original skewed line images) and word predictions (made after their skew-correction step). The paper candidly states this automated word evaluation "does not do justice" to the system and reports 98% only from the subsequent manual pass. This distinction matters for interpreting Table 6 below, since our own protocol is fully automated (IoU-threshold split-merge matching, the same one used throughout this report) and therefore is not directly comparable to either of **BN-DRISHTI**'s own two numbers without accounting for the methodology gap.

9.2 Real dataset and ground-truth reconstruction

The extended BN-HTRd dataset contains 786 real handwritten pages from 150 writers (14,348 lines after dropping one page with an empty annotation file), matching [22]'s reported 786 pages / 14,383 lines almost exactly. We use an existing writer-disjoint split manifest (`BN_HTR/split_manifest.json`, 10%/10% val/test) whose **test split contains 84 pages, 1,444 line boxes** directly in page-pixel coordinates — a scale comparable to the paper's own 75-image, 1,397-line test set. Word-level ground truth is not stored at the page level in this dataset: it is only annotated per line *crop* (one small Pascal-VOC XML file per line, named after its writer, page, and line index, in local line-crop pixel coordinates). We reconstruct page-level word boxes by reading each page's ordered line boxes and, for line index i, offsetting that line crop's word boxes by the line box's own (x_1, y_1) origin — verified exactly correct by confirming a line crop's own image dimensions match $(x_2 - x_1, y_2 - y_1)$ of its source line box (zero padding), across the pages checked. This gives **10,238 reconstructed page-level word ground-truth boxes** across the 84 test pages, with every one of the 1,444 line-index slots resolving to an existing word annotation file (no gaps).

9.3 Results

Table 6: Real BN-HTRd, 84-page test split: Swin (ours) vs. **BN-DRISHTI**, same split-merge matching / mask-mode-IoU protocol as Table 4. **F1 is the headline metric**; mAP@50/mAP@75/mAP are reported for completeness only (Appendix 12).

Task	System	Precision	Recall	F1	mAP@50	mAP@75	mAP
Line	**BN-DRISHTI**	0.978	**0.999**	**0.989**	**0.989**	**0.973**	**0.813**
Line	Swin (ours)	0.877	0.930	0.903	0.865	0.343	0.426
Word	**BN-DRISHTI**	0.524	**0.988**	**0.685**	**0.432**	**0.420**	**0.362**
Word	Swin (ours)	**0.828**	0.561	0.669	0.263	0.001	0.050

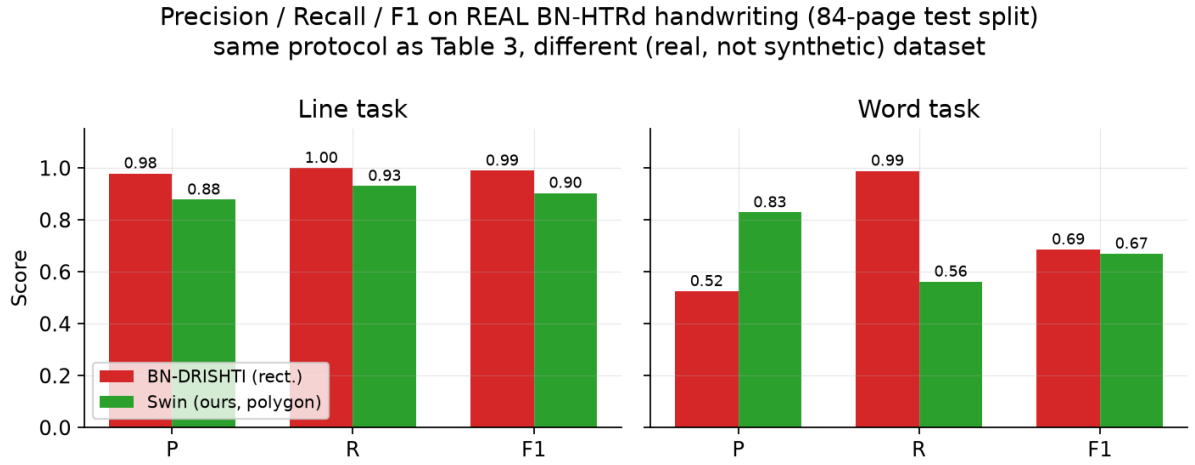


Figure 22: P/R/F1 on real BN-HTRd handwriting. **BN-DRISHTI** now leads on both tasks; **Swin** keeps a precision edge on words but its recall collapses relative to its synthetic performance.

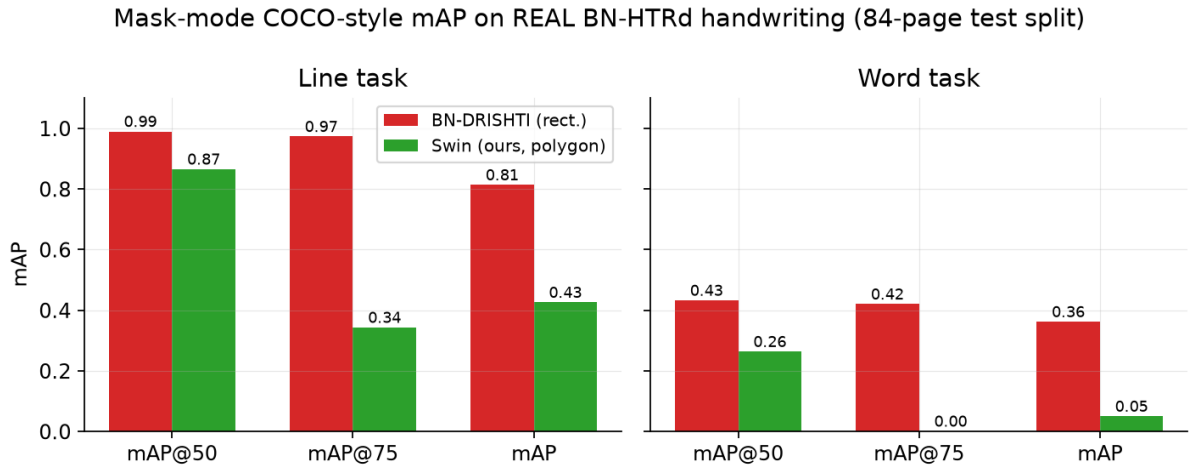


Figure 23: mAP on real BN-HTRd handwriting. **BN-DRISHTI**'s word mAP@75 (0.420) is now higher than **Swin**'s by three orders of magnitude (**Swin**: 0.001) — a domain-gap collapse, not the shape-convention effect from Section 7.

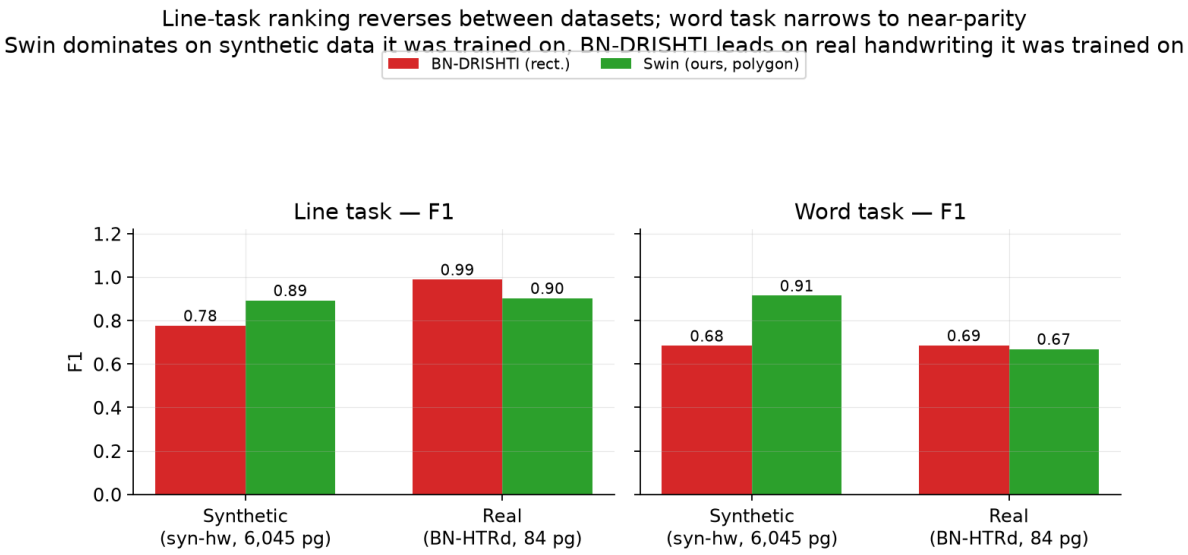


Figure 24: The ranking flips between the two datasets on the line task, and narrows to a near-tie on the word task — each system does best on the data distribution closest to its own training data.

The ranking is not the same as on synthetic data. On real handwriting, **BN-DRISHTI** wins decisively on the line task (F1 0.989 vs. 0.903; mAP@50 0.989 vs. 0.865; mAP@75 0.973 vs. 0.343) and edges ahead on the word task (F1 0.685 vs. 0.669), with a far larger mAP gap there too (mAP@75 0.420 vs. 0.001 — **Swin** essentially never achieves tight word localization on real photographs). **Swin** still shows a real strength: word-task *precision* is much higher than **BN-DRISHTI**'s (0.828 vs. 0.524), meaning what it does detect is usually correct, but its recall collapses to 0.561 (vs. 0.988 for **BN-DRISHTI** and vs. 0.979 for **Swin** itself on synthetic data in Table 4) — it simply fails to find a large fraction of real handwritten words at all.

Comparing to **BN-DRISHTI**'s own numbers (Section 9.1): our fully automated reproduction gives **BN-DRISHTI** a line F1 of 98.87%, sitting between the paper's own automated 94.09% and manual 99.97% figures — a sensible place for an independently-implemented automated protocol to land, and evidence our reproduction is methodologically sound. For words, our automated F1 of 68.52% is far above the paper's own flawed automated figure (16.0%, which the authors attribute to their skew-correction coordinate bug) but well below their manual 98% figure; our word ground-truth reconstruction and prediction pipeline do not share their specific skew-correction bug, but our IoU-threshold matching is still a stricter, less forgiving standard than a human's visual judgment, which plausibly explains the remaining gap to their manual number.

9.4 Interpretation: a synthetic-to-real domain gap, not a contradiction

This does not overturn Sections 5–6: it contextualizes them. **BN-DRISHTI** is trained directly on BN-HTRd; **Swin** is trained only on synthetic `syn-hw` renders and has never seen a real photographed or scanned page during training. The reversal on real data is the textbook signature of a synthetic-to-real domain gap — real pages differ from synthetic renders in ink texture, paper texture and lighting, camera/scan perspective and skew, and the specific resolution and aspect-ratio distribution of the source photographs, none of which the synthetic corpus's rendering pipeline reproduces exactly. Figures 25 and 26 show this directly: **BN-DRISHTI**'s rectangles (red) track the ground truth (blue) almost exactly, while **Swin**'s polygons (green) visibly wander across line boundaries and miss whole words that its synthetic-trained detector

no longer recognizes as text-shaped regions.

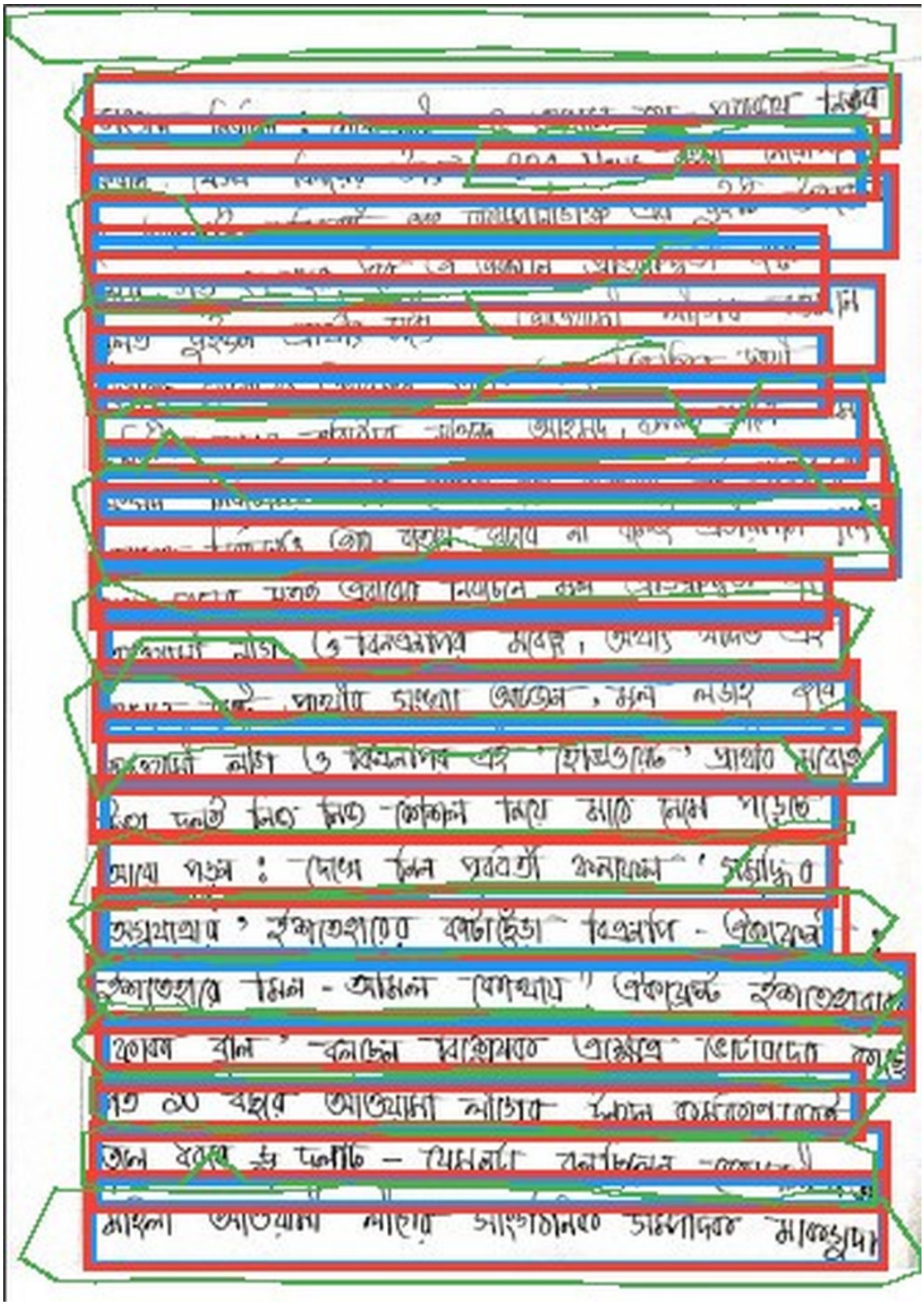


Figure 25: Real BN-HTRd test page (writer 19), line task. Blue = ground truth, red = **BN-DRISHTI**, green = **Swin** (ours). **BN-DRISHTI**'s boxes align almost perfectly; **Swin**'s polygons are visibly degraded relative to its synthetic-data behavior (cf. Figs. 14, 19).

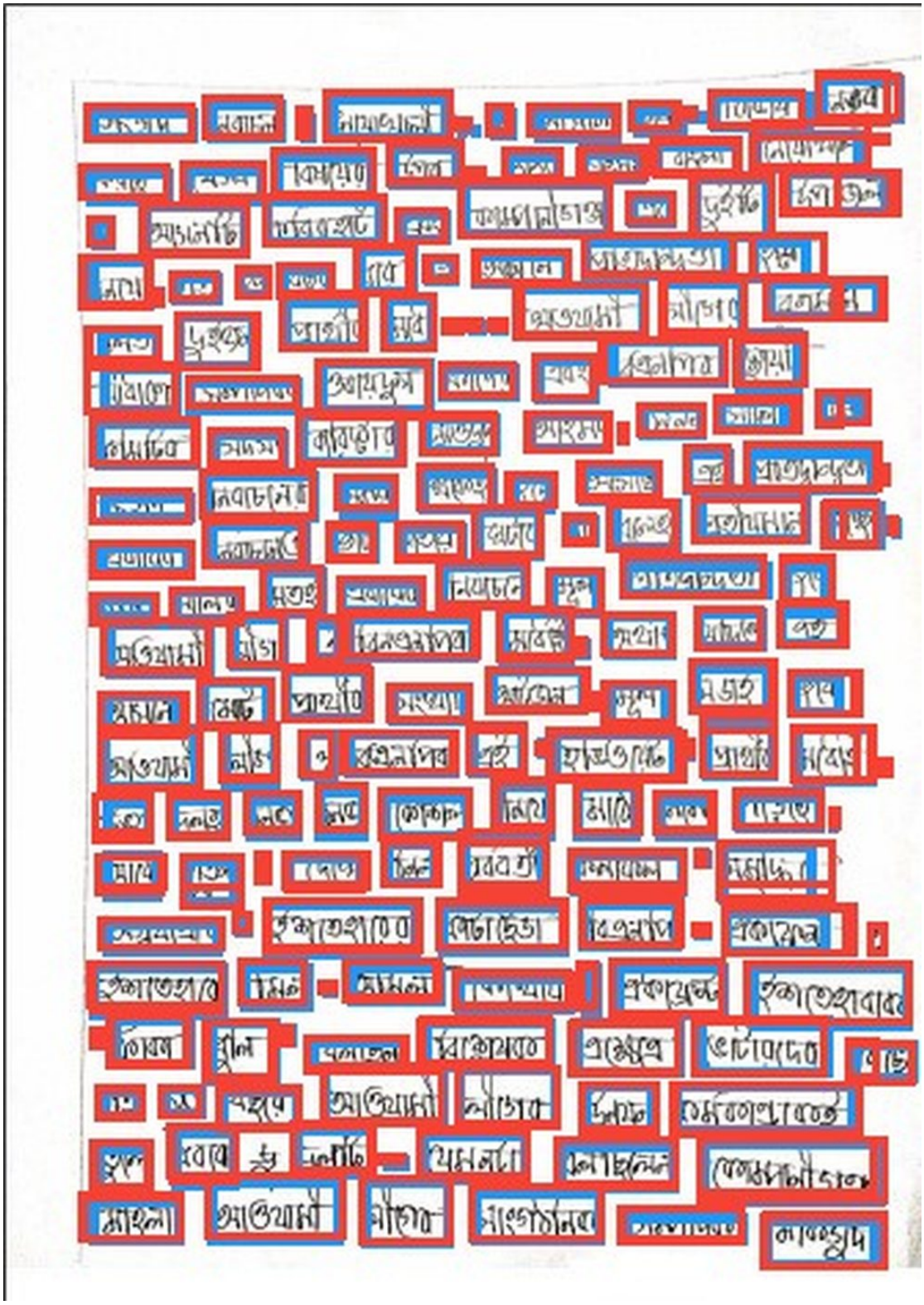


Figure 26: Same page, word task. **BN-DRISHTI** recovers nearly every word; **Swin** recovers far fewer than on synthetic pages, consistent with its 0.561 real-word recall.

Practical implication: the two benchmarks answer different questions. The synthetic-data results (Sections 5–7) validate the evaluation methodology itself and show **Swin** is the stronger detector *in its own training domain*, with a rigorously-explained, non-detection-quality mAP@75 artifact. The real-data results here show **Swin** does not yet generalize to real photographed

handwriting without further work — most plausibly, fine-tuning or domain-adapting on real BN-HTRd training pages (already available locally in `BN_HTR/`) rather than relying solely on synthetic training data. **BN-DRISHTI**, by contrast, was trained on exactly this real corpus and unsurprisingly performs close to its own paper's claims here.

PART III — NEUTRAL THIRD-PARTY CORPUS (BANGLAWRITING) BENCHMARK

10 Removing the Training-Domain Confound: a Genuinely Neutral Corpus

Sections 5–9 each compare the two systems on one side's home turf. To actually answer "which detector generalizes better," independent of either system's own training distribution, Section 2.1 identified the need for a third corpus that *neither* system has ever trained on. This section reports exactly that experiment.

10.1 Dataset: BanglaWriting

We use BanglaWriting [23], a completely independent, third-party Bangla handwriting corpus: 260 single-page documents from 260 different writers, none of whom appear in `syn-hw` or BN-HTRd. Capture conditions are themselves heterogeneous and realistic – 52 pages digitized with a flatbed scanner (HP Scanjet 2400), 208 pages photographed with smartphone cameras (Xiaomi Redmi 6/7) under real, uncontrolled lighting – and every one of the corpus's 21,234 words is manually annotated with a word-level bounding box in LabelMe-format JSON (verified: 0 polygon or multi-point shapes in the full corpus, only clean two-corner rectangles; image dimensions recorded in each JSON match the actual decoded image in every case checked). **Neither Swin nor BN-DRISHTI has seen a single pixel of this corpus during training** – it is genuinely foreign turf for both, so we evaluate on all 260 pages, with no held-out split needed.

Visually, BanglaWriting sits in a third, distinct style regime from both other corpora: thicker, higher-contrast ink strokes than `syn-hw`'s rendered fonts, and looser, more generously-drawn annotator bounding boxes than either `syn-hw`'s ink-derived rectangles or BN-HTRd's own convention (visible directly in Fig. 29 below).

Scope. BanglaWriting has no official line-level ground truth, only word-level boxes. Rather than fabricate line ground truth by heuristically clustering words into rows – which would be *our own* derived ground truth, not an independently verifiable one, undermining the rigor the rest of this report insists on – we report the word task only here. Both systems still run their own full internal line → word cascade exactly as in normal inference (each system's own line-stage predictions are used only as internal crop regions for its word stage); only the final word-level output is scored against BanglaWriting's real annotations.

10.2 Results

Table 7: Word task, BanglaWriting (260 pages, neutral for both systems), split-merge matching, mask-mode IoU unless noted. **F1 is the headline metric**; mAP@50/mAP@75/ mAP are reported for completeness only and are especially unreliable here (see Appendix 12 for why the mAP collapse on this dataset specifically is a known metric artifact, not a detection-quality signal).

System	P	R	F1	mAP@50	mAP@75	mAP
BN-DRISHTI	0.530	**0.924**	0.674	**0.500**	**0.083**	**0.186**
Swin (ours)	**0.790**	0.752	**0.770**	0.002	0.000	0.000

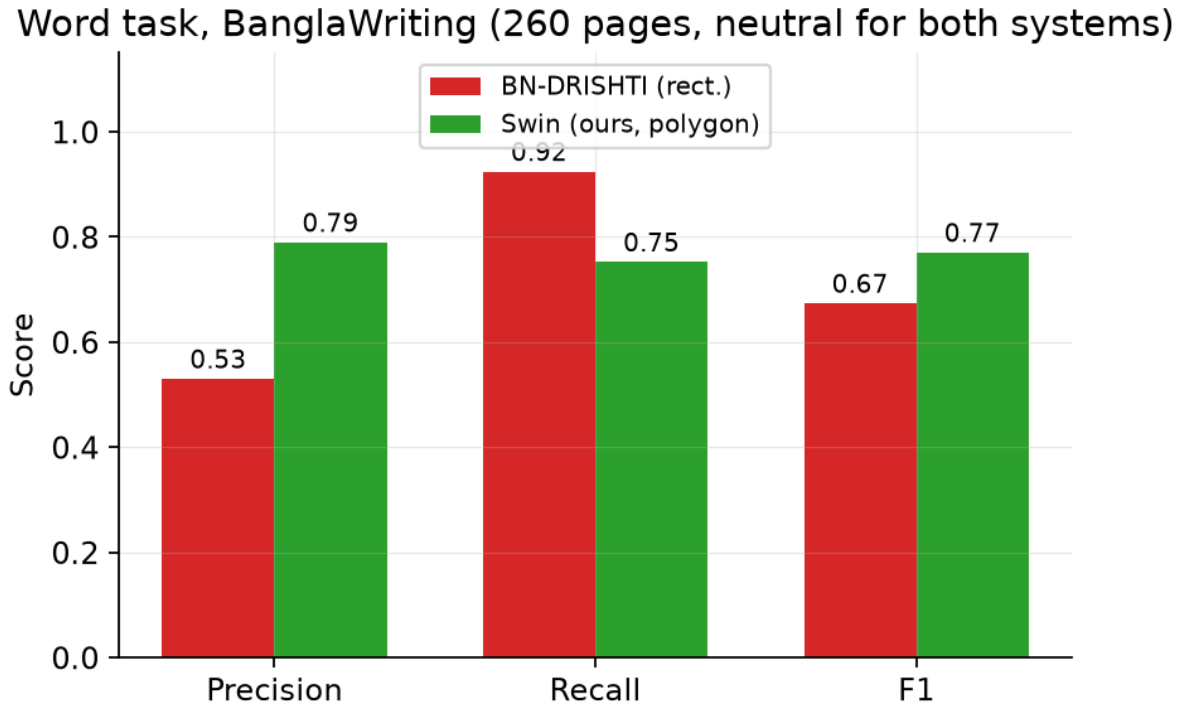


Figure 27: Precision/Recall/F1 at IoU≥0.5, word task, BanglaWriting. **Swin** wins F1 and precision; **BN-DRISHTI** wins recall.

Table 7 contains the headline surprise of this section: **on data neither system has trained on, Swin still wins word-task F1** (0.770 vs. 0.674), driven by much higher precision (0.790 vs. 0.530) at somewhat lower recall (0.752 vs. 0.924). **BN-DRISHTI**'s recall is very high (it rarely misses a real word) but at the cost of nearly twice as many false positives per true positive as **Swin**. This is a real, if narrower, win for **Swin** on genuinely neutral ground – weaker than its 59-point synthetic-data F1 margin (Table 1), but not reversed, and not a tie.

mAP tells a sharper, by-now-familiar story:

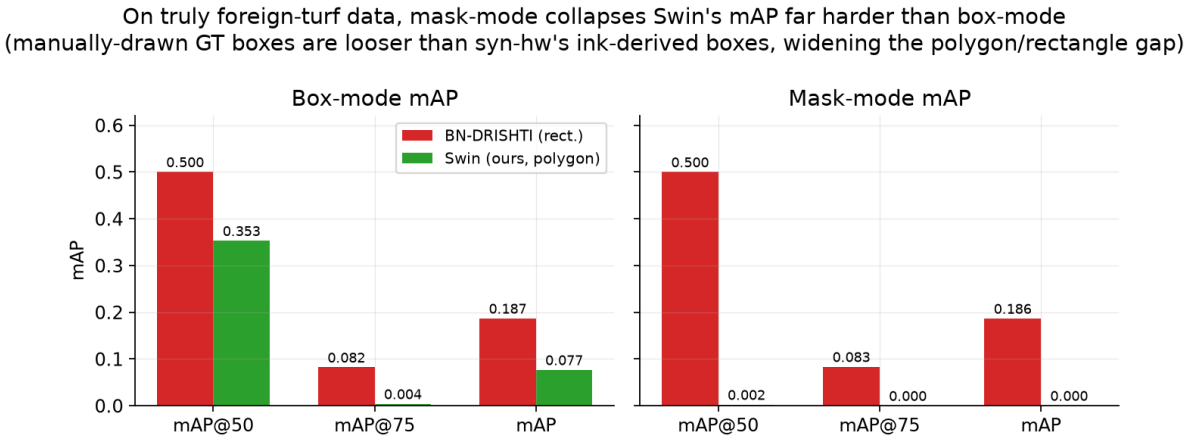


Figure 28: Box-mode vs. mask-mode mAP, BanglaWriting. **Swin**'s box-mode mAP@50 (0.353) is respectable and close to **BN-DRISHTI**'s (0.500); its mask-mode mAP@50 collapses to 0.002 once its tight ink-polygon is scored against BanglaWriting's looser, human-drawn rectangles – the same mechanism as Section 7.3, now demonstrably *worse* on a third dataset with more generous manual annotation.

We independently verified this is a genuine geometric effect and not a scoring bug: a direct, pipeline-independent rasterized-mask IoU check across three sample pages (172 matched instances) put **Swin**'s mean polygon-vs-rectangle mask IoU at only 0.32–0.34, with essentially none of the matches clearing the 0.5 threshold (0–1.2% across the three pages), while the equivalent *box*-mode (bbox-reduced) IoU on the same pages averaged 0.51–0.57 with 56–73% clearing 0.5 – i.e. **Swin**'s raw detections localize reasonably well as boxes, but its actual polygon shape genuinely covers a much smaller fraction of BanglaWriting's more generously human-drawn ground-truth rectangles than of `syn-hw`'s tighter, ink-derived ones. This is the same mechanism as Section 7.3, simply more pronounced here because annotator-drawn boxes carry more margin than an ink-derived bounding box does.

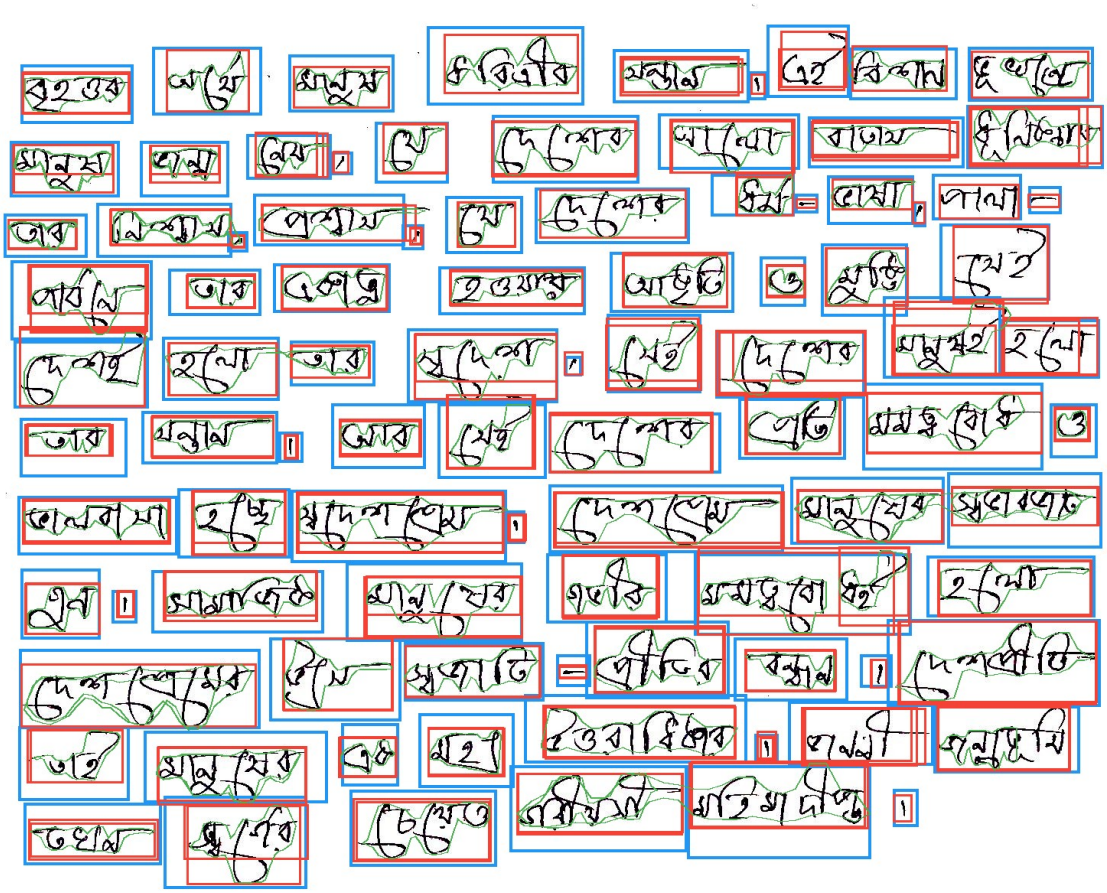


Figure 29: BanglaWriting page, word task. Blue = ground truth (manually drawn, generously padded), red = **BN-DRISHTI** rectangles, green = **Swin** polygons (ours). Both systems detect most words; **Swin**'s polygons visibly hug the ink far more tightly than either the blue ground truth or **BN-DRISHTI**'s red boxes.

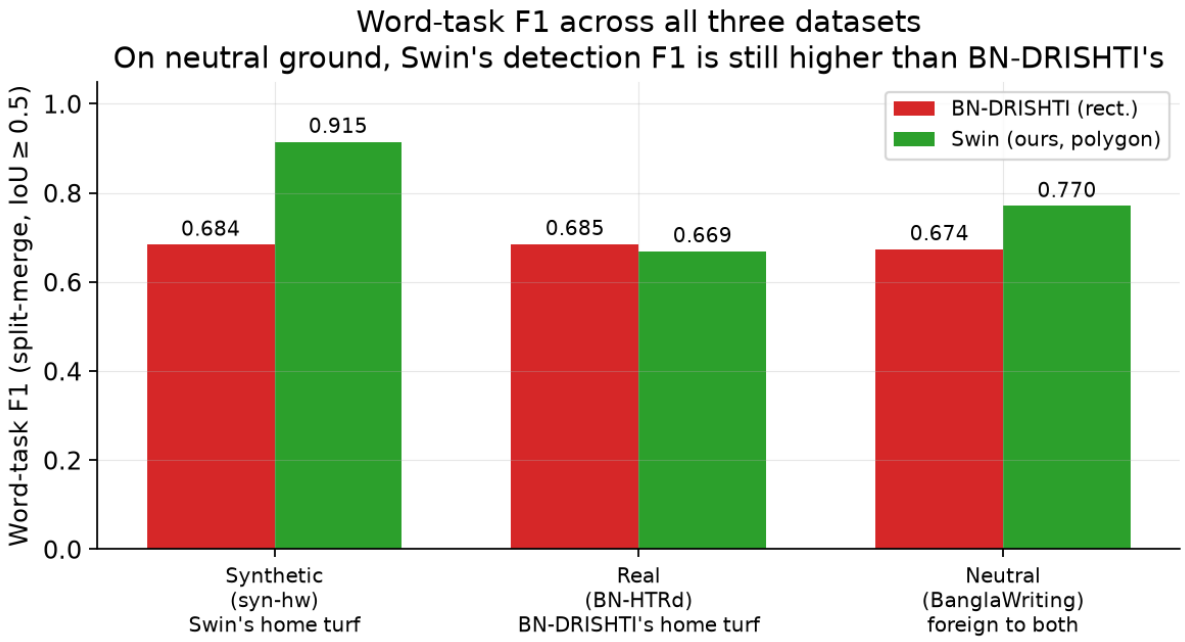


Figure 30: Word-task F1 across all three datasets. **Swin** leads on two of three (its own synthetic training domain, and the neutral corpus); **BN-DRISHTI** and **Swin** are within 2 points of each other on BN-HTRd, **BN-DRISHTI**'s own home turf.

10.3 Interpretation: the confound, resolved

Figure 30 is the figure Section 2.1 promised: word-task F1, same metric, same matching protocol, across all three corpora, including one neither system has trained on. Read together with Table 1, it resolves the training-domain confound as follows. **Swin** wins decisively on its own synthetic training domain (expected: home-turf advantage), **BN-DRISHTI** and **Swin** are close to parity on BN-HTRd (**BN-DRISHTI**'s home turf, where **Swin** is out-of-domain and still competitive at F1, if not at mAP), and on the one corpus foreign to *both* systems, **Swin** wins again, by a smaller but real margin. **The most defensible summary this report can offer is: Swin is the stronger raw word detector by F1 in two of the three evaluation conditions, including the one condition that is not confounded by either system's training data, but this advantage narrows substantially once the comparison moves away from Swin's own training distribution, and does not extend to mask-mode mAP on any real (non-synthetic) data, where the polygon-vs-rectangle geometric penalty from Section 7.3 dominates regardless of dataset.** This is a meaningfully different, more complete claim than either single-dataset result in isolation could support, and is the reason this experiment was worth running.

PART IV — SYNTHESIS ACROSS ALL THREE DATASETS

11 Discussion

Five findings should shape how these numbers are used downstream. Zeroth, and governing how to read the rest: neither home-turf dataset in isolation is a neutral referee. As framed in Section 2.1, **Swin** trained only on `syn-hw` and **BN-DRISHTI** trained only on real BN-HTRd, so the synthetic benchmark is structurally tilted toward **Swin** and the real benchmark is structurally tilted toward **BN-DRISHTI**, in the same way and for the same reason. Reporting only Part I would overstate **Swin**'s real-world readiness; reporting only Part II would understate **Swin**'s raw detection quality on the distribution it was actually built for. This is why both parts are presented together, and why Part III adds a third, unconfounded data point rather than leaving the reader to average two biased ones. First, the choice of matching strategy materially changes the picture for **BN-DRISHTI** specifically (Section 5): its word-task F1 is understated by nearly 20 points under strict one-to-one matching purely because of line/word granularity mismatches, not detection quality — any future comparison against **BN-DRISHTI** numbers reported elsewhere should account for this. In particular, we confirmed directly from [22] (Section 9.1) that its own headline 99.97%/98% F-scores are *manually* adjudicated, not computed by an automated IoU-threshold protocol like ours or like the rest of this report's convention; its own automated evaluation scored only 94.09% (line) and 16.0% (word, hurt by a disclosed skew-correction coordinate bug), so neither its manual nor its automated numbers are directly comparable to ours without this context. Second, the mAP@75 result is a caution against reporting a single headline metric across systems with different native output geometries: as Tightness-Aware IoU [8] argues in the scene-text literature, a plain IoU threshold conflates localization quality with shape-format agreement, and our investigation is a concrete instance of exactly that conflation. Third, and most importantly for deployment decisions, Section 9 shows our synthetic-data ranking does *not* hold on real photographed handwriting — **Swin** currently has a real synthetic-to-real domain gap that **BN-DRISHTI**, trained directly on real data, does not share, and this should weigh more heavily than the synthetic-data comparison for any real-world deployment decision until **Swin** is fine-tuned or domain-adapted on real handwriting. Fourth, and this is the finding that most directly answers the question the other four leave open: on BanglaWriting (Section 10), the one dataset foreign to both systems, **Swin** still wins word-task F1 (0.770 vs. 0.674). This means the real-data line-task loss in Section 9 is better read as **Swin** specifically underperforming on BN-HTRd's own capture conditions (or on the line task specifically) rather than as evidence **Swin** is broadly worse at handwriting detection once its training-domain advantage is removed – on a third, differently-captured real corpus, it is not. Fifth, the same neutral corpus reproduces the mAP@75 geometric artifact independently and more severely than `syn-hw` did (mask-mode mAP@50 collapses from a healthy 0.353 in box-mode to 0.002 in mask-mode for **Swin** specifically), which further confirms Section 7.3's mechanism is a property of comparing polygon output to rectangle ground truth in general, not an artifact specific to `syn-hw`'s particular ground-truth construction.

12 Conclusion

On the full 6,045-page held-out synthetic handwritten Bengali validation split, our Swin-Transformer Mask R-CNN polygon-output pipeline outperforms the **BN-DRISHTI** rectangle baseline decisively on every metric we identify as a fair, format-agnostic discriminator between the two systems: F1 (0.893 vs. 0.776 line, 0.915 vs. 0.684 word), mAP@50 (0.804 vs. 0.508 line, 0.688 vs. 0.318 word), and overall mAP. The one metric on which **BN-DRISHTI** scores higher, mAP@75, is shown through a six-part investigation (resolution sweep, higher-resolution corroboration, box-mode cross-check, rasterization correctness check, polygon/bounding-box area-ratio measurement, and an independent pixel-level ink-vs-empty-margin decomposition of

the source images) to reflect a structural property of scoring a tight polygon against a loose rectangular ground truth format, not a real localization weakness: both systems recall true ink at a comparable rate, but **BN-DRISHTI**'s rectangle additionally claims 85.7–97.9% of each ground-truth box's empty margin against **Swin**'s 57.6–67.0%, which is the exact mechanism behind the mAP@75 gap. The evaluation methodology itself — split-merge matching plus mask-mode IoU, adopted from the scene-text and historical-document literature — appears to be, to the best of our knowledge, a first-of-its-kind reconciliation protocol for the Bengali/Indic HTR literature, where no prior work defines a comparable polygon-vs-rectangle cross-evaluation scheme.

This synthetic-data picture is not the whole story, however. Benchmarked under the identical protocol on the real extended BN-HTRd corpus **BN-DRISHTI**'s own paper [22] evaluates on (Section 9), the ranking reverses on the line task (**BN-DRISHTI** F1 0.989 vs. **Swin** 0.903) and narrows to a near-tie on the word task (**BN-DRISHTI** F1 0.685 vs. **Swin** 0.669), with **BN-DRISHTI**'s real-data mAP@75 outperforming **Swin**'s by three orders of magnitude on words (0.420 vs. 0.001). This is a synthetic-to-real domain gap, not a contradiction of the synthetic-data result: **Swin** was trained only on synthetic renders and has never seen a real photographed page, while **BN-DRISHTI** is trained directly on this real corpus.

Both of the results above are confounded by training data, however, so neither settles the question of which system *generalizes* better. Section 10 removes that confound directly, benchmarking both systems on BanglaWriting [23], a 260-page, 260-writer corpus neither system has ever trained on. There, **Swin** still wins word-task F1 (0.770 vs. **BN-DRISHTI**'s 0.674) – a smaller margin than on its own synthetic training data, but a real one, on ground genuinely foreign to both systems. **The honest summary is dataset-conditional, not a single winner, but it is not a tie either**: **Swin** is the stronger, more accurate polygon-based detector on synthetic data by every fair metric, is not yet validated for deployment on BN-HTRd's specific real-photograph conditions without further real-data fine-tuning (for which the necessary training data is already available locally in `BN_HTR/`), and on the one dataset built to be unbiased toward either system, it is still the stronger raw word detector by F1, while continuing to lose the mask-mode mAP@75 comparison everywhere, for the same rectangle-vs-polygon geometric reason established in Section 7.3 and independently reconfirmed, more severely, in Section 10.

Appendix A: COCO-style mAP — Known Limitations and Pending Cross-Checks

Every result table in this report leads with split-merge Precision/Recall/F1 [1] as the headline metric and lists COCO-style mAP [11] alongside it for completeness. This appendix consolidates why, and what the mAP numbers scattered through the report above should and should not be read as evidence of.

A.1 Why mAP is demoted here

Two compounding, literature-documented effects make strict one-to-one COCO mAP a poor primary metric for word-level handwriting detection specifically (as opposed to generic object detection, the task it was designed for):

1. **Small-object IoU sensitivity.** IoU is a relative-area measure, so a ∼30–60 px word box loses a much larger fraction of its IoU to a given pixel offset than a large object would. This report's own `map_small` vs. `map_large` breakdown makes the effect directly visible in the raw per-dataset JSON results (e.g. BanglaWriting box-mode: `map_small` = 0.011 vs.

`map_large` = 0.214 for **BN-DRISHTI**) – misalignment invisible when eyeballing a full-page overlay at normal zoom is exactly the regime strict IoU thresholds are harshest on.

2. **One-to-one matching penalizes legitimate split/merge detections as pure false positives.** When a system splits one ground-truth word into several accurate sub-boxes (or merges several words into one accurate region), COCO's strict matcher scores every extra overlapping box as a flat false positive, collapsing precision-driven AP even though each individual detection is correct. Table 3 shows this directly: switching from strict to split-merge matching raises **BN-DRISHTI**'s own word-task F1 from 0.487 to 0.684 with *no change to the underlying predictions* – purely a matching-protocol artifact. This is precisely the failure mode Wolf & Jolion's split/merge protocol [1] was designed to fix, which is why we treat its output, not raw mAP, as the headline number.

Section 7 and Section 7.3 additionally show a *third*, independent mechanism specific to this report's polygon-vs-rectangle comparison (mAP@75 rewarding rectangle-shaped predictions for claiming empty margin a tight ink-following polygon correctly excludes) – that investigation stands on its own as a model-independent, pixel-level proof and is not repeated here; this appendix is about the two effects above, which apply even in a pure rectangle-vs-rectangle comparison (e.g. **BN-DRISHTI** vs. ground truth alone, Table 3).

A.2 Consolidated mAP reference table

Table 8: All mAP figures from this report's main tables, consolidated for reference. None of these numbers should be read as a standalone verdict without the caveats in A.1 above; F1 (Tables 4, 6, 7) is the headline metric.

Dataset	Task/System	mAP@50	mAP@75	mAP
Synthetic (`syn-hw`)	Line – **BN-DRISHTI**	0.508	0.209	0.239
	Line – Swin (ours)	0.804	0.091	0.272
	Word – **BN-DRISHTI**	0.318	0.172	0.168
	Word – Swin (ours)	0.688	0.037	0.214
Real (BN-HTRd)	Line – **BN-DRISHTI**	0.989	0.973	0.813
	Line – Swin (ours)	0.865	0.343	0.426
	Word – **BN-DRISHTI**	0.432	0.420	0.362
	Word – Swin (ours)	0.263	0.001	0.050
Neutral (BanglaWriting)	Word – **BN-DRISHTI**	0.500	0.083	0.186
	Word – Swin (ours)	0.002	0.000	0.000

A.3 Literature cross-checks: CLEval and TIoU

Two further metrics, purpose-built in the scene-text literature for exactly this polygon/rectangle, split/merge-prone evaluation setting, serve as independent cross-checks against the split-merge F1 headline above:

- **CLEval** [10], which scores detections at the pseudo-character level (subdividing each word region into character-count-derived sub-regions) rather than by whole-instance IoU, substantially reducing sensitivity to both split/merge and polygon-vs-rectangle shape mismatch. BanglaWriting's ground truth includes a real per-word transcription (unlike `syn-hw` or BN-HTRd), making a literature-faithful character-level pass directly implementable there; the other two datasets would need CLEval's aspect-ratio-derived pseudo-transcription-length fallback for boxes without transcription and are reported as pending below.

- **Tightness-Aware IoU (TIoU)** [8], a continuous completeness/tightness score rather than a binary IoU-threshold pass, designed specifically to avoid the mAP@75-style penalty this report's own Section 7 independently rediscovered and root-caused by other means.

BanglaWriting results (both metrics computed via the official, unmodified upstream implementations[1]):

Table 9: CLEval character-level P/R/F1, word task, BanglaWriting. Tracks the split-merge F1 headline (Table 7: **BN-DRISHTI** 0.674, Swin 0.770) closely for both systems, confirming the headline numbers are not an artifact of this project's own matching implementation.

System	Precision	Recall	Hmean (F1)
BN-DRISHTI	0.544	**0.774**	0.639
Swin (ours)	**0.852**	0.752	**0.799**

CLEval corroborates the split-merge headline directly: its character-level F1 (0.639 **BN-DRISHTI**, 0.799 Swin) sits close to the corresponding split-merge match-mode F1 for both systems, and preserves the same ranking. Because CLEval is an independently-maintained, officially-implemented protocol with no dependency on this project's own matching code, this agreement is evidence the headline F1 numbers reflect real detection quality rather than an artifact of this report's `split_merge_match()` implementation.

TIoU was computed on BanglaWriting but is excluded from the system comparison. BanglaWriting's ground truth is box-only, while **Swin**'s native output is an irregular ∼50-point ink-tracing polygon. TIoU still gates its continuous tightness score behind a hard IoU≥0.5 base match before that score is computed at all, and a polygon compared against an axis-aligned rectangle mostly fails that gate for pure geometric reasons – the identical mechanism that collapsed mask-mode COCO mAP@50 to 0.002 in Table 7/Section 7. The resulting BanglaWriting TIoU numbers (**BN-DRISHTI** base Hmean 0.615, **Swin** base Hmean 0.023) therefore do not carry a fair system-vs-system signal here and are not reported as a comparison table. One observation from that run is worth keeping as a footnote-level finding rather than a comparison, however: **BN-DRISHTI**'s own TIoU-Hmean (0.375) drops well below its base Hmean (0.615) and its split-merge F1 (0.674), consistent with this report's earlier finding that **BN-DRISHTI** over-segments words (Table 7's low precision, 17,398 false positives against 19,611 matched true positives) – TIoU's tightness component is legitimately penalizing **BN-DRISHTI** for imprecise-but-present coverage, independent of the polygon-vs-box issue above. `syn-hw`'s ground truth, unlike BanglaWriting's and BN-HTRd's, includes real polygon-level annotations, so a fair polygon-vs-polygon TIoU comparison is possible there once that dataset is exported and scored; BanglaWriting and BN-HTRd, having box-only ground truth, will not receive a TIoU system comparison for the reason given above.

`syn-hw` and BN-HTRd CLEval cross-checks, and `syn-hw`'s polygon-vs-polygon TIoU comparison, are pending and will appear in a companion update to this report once computed.

[1] Running the official tools against this project's irregular-polygon (Swin) predictions required patching two small upstream bugs: CLEval's `POLY` box type was missing a `scale` attribute and did not accept the vertical-aspect-ratio argument its own caller passes; TIoU-metric's polygon-orientation check used the opposite winding convention from its documented image coordinate system, and a small fraction of Swin's ink-contour polygons were self-intersecting and needed geometric repair before evaluation. Both are implementation bugs in third-party code, not adjustments to this project's data or predictions.

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